

ANNALES ACADEMIÆSCIENTIARUM FENNICÆ

MATHEMATICA

DISSERTATIONES

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ELECTRICAL IMPEDANCE TOMOGRAPHY
AND FADDEEV GREEN'S FUNCTIONS

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Dissertation for the Degree of Doctor of Technology to be presented
with due permission for public examination and debate
in Auditorium N at Helsinki University of Technology (Espoo, Finland)
on the 26th of November, at 12 o'clock noon.

HELSINKI 1999
SUOMALAINEN TIEDEAKATEMIA

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Academia Scientiarum Fennica
ISSN ??????
ISBN ??????

Received 4 October 1999

1991 Mathematics Subject Classification:
35J25,35R30,78A70,65R10,65R20.

YLIOPISTOPAINO
HELSINKI 1999

Acknowledgements

I am grateful to Erkki Somersalo for being the advisor of my work and for introducing me to the international inverse problems research community. Also, our combined boxing and mathematics sessions with him and Lassi Päiväranta were extremely inspiring.

I thank Petri Ola and Margaret Cheney for carefully reading my manuscript and giving valuable suggestions; I also thank Rainer Kress for being my opponent. I am grateful to Adrian Nachman for his interest to this work and the enlightening discussions about his results.

This work led to wonderful collaboration with Jennifer Mueller and David Isaacson. Thanks to them together with Jon Newell and Margaret Cheney, my stay at the Rensselaer Polytechnic Institute was very pleasant. I appreciate the opportunity to be measured with the ACT 3 impedance imaging device (it didn't hurt).

I thank Gunther Uhlmann for discussions about the inverse conductivity problem and John Sylvester for his advice in the numerical aspects of my work.

I wish to thank my dear friend Matti Lassas for always having time for my questions, mathematical and others. I thank all the people in the Finnish inverse problems research society for creating a great social atmosphere. I thank Eero Saksman, Hans-Olav Tylli and Juha Kinnunen for their help in questions related to singular integrals and functional analysis.

I am grateful to my wife and friends for supporting me during this work.

Finally, I thank Olavi Nevanlinna for arranging an excellent working environment at the Helsinki University of Technology and the Finnish Ministry of Education, Finnish Academy of Science and Letters, the Emil Aaltonen Foundation, the Instrumentarium Science Foundation and the Finnish Cultural Foundation for financial support.

Tuusula, October 1999

Samuli Siltanen

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1. Introduction

1.1. Electrical impedance tomography

Magnetic resonance imaging (MRI and fMRI), X-ray tomography or computed tomography (CT), magnetic encephalography (MEG) and ultrasound imaging are examples of medical imaging methods that produce a picture of a patient's inner organs without intruding into the body. Images coming from different systems complement each other since each of these methods measures the distribution of some specific physical parameter, such as mass density (CT) or sound speed (ultrasound imaging).

Some of the methods are more harmful to the patient than others: X-ray measurements expose her to dangerous radiation whereas ultrasound imaging has little or no health risks. Another difference between the systems is that some of the machines are very expensive. For these reasons it is valuable to have as many different imaging methods available as possible.

Electrical impedance tomography (EIT) is a medical imaging system that produces a picture of electric conductivity distribution inside the patient. For the EIT measurement an array of electrodes is attached for instance around the chest of the patient as shown in Figure 1. Measurements are done by feeding electric currents

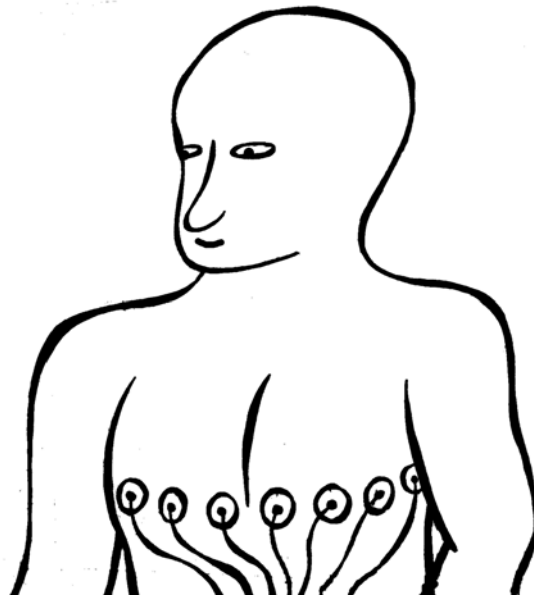


Figure 1: EIT measurements can be done with an array of electrodes around the chest.

through the electrodes and measuring the corresponding voltages at the electrodes. This measurement is repeated with several different choices of current patterns. The

resulting image is an approximate map of the electric conductivity in a horizontal slice on the level of heart and lungs.

To get a feeling about why this should be possible, note that if the lungs are full of air, no electric current will pass through them. On the other hand, empty lungs do conduct electricity to a somewhat greater extent, so at least these two situations will give different measurements. Of course it is a lot more challenging task to reconstruct every detail inside the chest; this requires precise measurements and nontrivial mathematics.

Measuring around the chest can be used in monitoring the heart or to improve ECG measurements. For other medical purposes of EIT, such as screening for breast cancer, see [24]. Attractive features of EIT include the possibility of harmless monitoring and good temporal resolution. Moreover, EIT gives complementary information to other imaging modalities.

There are working EIT machines, most notably the Adaptive Current Tomography (ACT) system built in Rensselaer Polytechnic Institute (RPI). It is capable of real-time imaging and has been tested on people. See [11] for more details. Impedance imaging is also used in a commercial breast cancer detector called T-Scan that was launched recently.

EIT is not yet in wide clinical use because practically reliable and fast imaging is extremely difficult and needs still more research. This work is part of a project aiming to produce a new imaging algorithm for EIT starting from the latest results of applied mathematics. Hopefully this will help the ongoing effort to build accurate, cheap and portable EIT devices.

1.2. Mathematics of EIT

The mathematical task of interpreting EIT measurements is an inverse boundary-value problem called the *inverse electric conductivity problem*. It is a challenging question in the theory of partial differential equations since it is nonlinear (Figure 2) and ill-posed. Designing a practical algorithm for EIT is hard because nonlinearity makes the mathematical solution theoretically difficult and ill-posedness means that a small error in the practical measurements may cause a large error in the reconstructed image. Due to its challenging theoretical nature and medical applications, EIT is a good problem in applied mathematics.

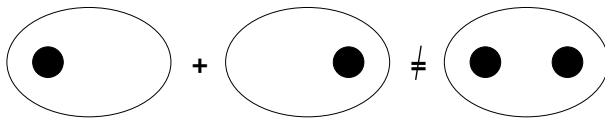


Figure 2: EIT is a nonlinear problem. The boundary measurements corresponding to the two leftmost objects do not add up to the measurement corresponding to the rightmost object.

Let us state the inverse conductivity problem in mathematical terms: let $\Omega \subset \mathbb{R}^2$ be a bounded, simply connected Lipschitz domain and $\gamma : \Omega \rightarrow \mathbb{R}$ a bounded, measurable real function with a strictly positive lower bound, that is, $0 < c \leq \gamma(x)$ a.e. The *forward* electric conductivity problem is to determine, given any $f \in H^{1/2}(\partial\Omega)$, the H^1 solution u of the Dirichlet problem

$$\begin{cases} \nabla \cdot \gamma \nabla u &= 0 \text{ in } \Omega, \\ u|_{\partial\Omega} &= f. \end{cases} \quad (1)$$

Note that the boundary value problem (1) is elliptic due to the lower bound of γ and its solution under various assumptions on γ and $\partial\Omega$ is discussed in [17, 20]. Define now the Dirichlet-to-Neumann map of γ by

$$\Lambda_\gamma : H^{1/2}(\partial\Omega) \rightarrow H^{-1/2}(\partial\Omega), \quad \langle \Lambda_\gamma f, g \rangle = \int_\Omega \gamma \nabla u \cdot \nabla v,$$

where v is any $H^1(\Omega)$ function with trace g on the boundary. The *inverse* electric conductivity problem is to find out if γ (satisfying perhaps some additional regularity assumptions) is uniquely determined from the knowledge of Λ_γ and if so, to reconstruct γ .

Physically f represents a voltage distribution on the boundary of a two-dimensional physical body Ω with an unknown inhomogeneous conductivity distribution γ . Furthermore, u is the electric potential inside the body that solves the conductivity equation (1) and agrees with f on the boundary. Thus the Dirichlet-to-Neumann map Λ_γ can be interpreted as the voltage-to-current map

$$\Lambda_\gamma f = \gamma \frac{\partial u}{\partial \nu} \Big|_{\partial\Omega}$$

taking the boundary value of the voltage to the resulting electric current through the boundary. In this way Λ_γ represents static electric measurements done on the boundary of Ω and the inverse conductivity problem is to determine the conductivity that caused those measurements.

Let us make some observations:

- The physical significance of the assumption that the conductivity is bounded away from zero is that there cannot appear annulus-shaped areas of zero conductivity; this would make it impossible to reconstruct anything inside the annulus.
- As mentioned above, in practical measurements one feeds currents and measures voltages and not the other way around. This is due to the fact that although the two measurements are mathematically equivalent, the latter procedure is unstable to measure [18].

- Although physical bodies are three-dimensional, from the point of view of EIT the problem of producing the picture of a horizontal slice of a person's chest is approximately two-dimensional. This is why we consider only the two-dimensional case in this work. For an analysis of avoiding this simplification by using approximate Dirichlet-to-Neumann maps, see [33].
- In addition to EIT, there are other physical applications of the inverse conductivity problem, such as geological prospecting and non-destructive material testing. All further references to applications will, however, concentrate on EIT.

1.3. A brief history of the problem

Research on the inverse conductivity problem has two traditions: a purely mathematical one providing uniqueness and reconstruction results and a practical approach having as its goal algorithms that perform well on physical data.

The theoretically oriented line of research was started by Calderón [8], who studied electrical engineering before changing to mathematics. He stated the inverse conductivity problem for the first time in the above form (in dimensions $n \geq 2$) and gave some preliminary results for conductivities close to a constant. Mathematically his work continued the tradition of Gel'fand, Levitan and Börg [16, 6].

Once Calderón asked his question, new results started to appear. Kohn and Vogelius [35] showed that for piecewise analytic conductivities the Dirichlet-to-Neumann map uniquely determines the conductivity in dimensions 2 or more. Sylvester and Uhlmann [50] showed in dimensions $n \geq 3$ that if $\partial\Omega$ is C^∞ then Λ_γ uniquely determines $\gamma \in C^\infty(\overline{\Omega})$.

The first reconstruction method was published by A. Nachman [42]; this result is valid in dimensions strictly greater than two. The work of R. G. Novikov [23] contains similar results.

Meanwhile the case $n = 2$ remained unsolved. Why is the two-dimensional case more difficult than the higher-dimensional ones? A naive reason for this is that in n dimensions the Schwartz kernel of Λ_γ depends on $2(n - 1)$ variables while γ is a function of n variables; this means that in a sense there is no overdeterminacy in two dimensions and all the data has to be used for the reconstruction.

In 1996 A. Nachman published a reconstruction procedure for the two-dimensional problem [43]. This method is especially interesting because it is constructive in nature. The smoothness assumption in [43] is $\gamma \in W^{2,p}(\Omega)$ for some $p > 1$ which implies that γ is continuous. This result was soon sharpened for conductivities in $W^{1,\lambda}(\Omega)$, $2 < \lambda < \infty$, by R. Brown and G. Uhlmann [7]. These conductivities are also continuous.

A more detailed introduction to the mathematical development of the problem is given in [43, 51]. Also V. Isakov's article [30] and book [31] are good references;

they cover also results aiming at partial answers such as finding domains with zero or infinite conductivity. See also [26, 27].

The practical branch of EIT research started with papers like [22], although the inverse conductivity problem has been treated earlier in the engineering literature [38, 9].

Since the inverse conductivity problem is very complicated mathematically, we can expect the the practical EIT imaging task to be difficult. In addition to the nonlinearity and ill-posedness of the theoretical model there are further problems: the Dirichlet-to-Neumann map is not given; instead its inverse operator has to be approximated from noisy and finite-precision current-to-voltage measurements. Furthermore, only a finite number of current patterns can be applied through a finite number of electrodes. This situation places restrictions on the resolution of the images [29] and requires the physical model of the electrode system to be refined [46]. This leads to the notion of *distinguishability* [28] describing the resolution it is possible to obtain in principle with a given system.

With these restrictions known, many algorithms have been developed and tested on real data. They fall roughly in three categories:

1. Solve the linearized inverse problem. This approach works best for conductivities close to constant; examples include back-projection [3] and one-step Newton schemes like NOSER [14].
2. Solve the nonlinear problem by finding iteratively a conductivity distribution that matches the physical measurements best in the least squares sense. These methods require regularization due to the ill-posedness of the inverse problem. For a list of references to methods of this type see [11, p.90]; note also [12, 34]. See [55, 56] for reports on using anatomical a priori information in iterative EIT imaging.
3. Solve the problem directly. The methods of this category are called direct methods as opposed to iterative ones; they aim as well to solve the full nonlinear problem. An existing example of this type is layer stripping [47, 49]. The numerical realization of Nachman's reconstruction method will fall into this category.

Some of these ideas combined with nontrivial electrical engineering make it possible to build reasonably working EIT devices. This has been done in several laboratories around the world, e.g. RPI, Sheffield, Oxford and Kuopio, Finland, see [11, 39, 56]. We mention that the ACT system of the RPI group performs imaging in real time and has been tested in a clinical environment. For ideas for dynamical measurements using Kalman filters see [57].

The above treatment of the practical aspects of EIT is far from complete. For a more thorough review see [59, 10, 11].

1.4. Applying A. Nachman's method to EIT

The two main lines of EIT research, the purely mathematical and the applied one, have not benefited much from each other. The methods used by the theoretical reconstructions tend to be hard or impossible to realize on a computer. On the other hand it has turned out to be difficult to prove convergence results for the existing practical algorithms; in addition it is not easy to describe the relation between the actual conductivity and a reconstruction computed with a convergent but regularized algorithm.

Nevertheless, Nachman's mathematical proof [43] for the two-dimensional case is throughout constructive and seems not to have steps that are impossible to implement practically. This work is part of a project to develop a practical reconstruction algorithm based on Nachman's method, the scope being to provide both theoretical insight and numerical algorithms for various steps of the reconstruction. In this way this thesis helps to build a bridge between theory and practice of the inverse conductivity problem.

The larger project of realizing Nachman's whole reconstruction scheme is carried out by the author in collaboration with professor David Isaacson and PhD Jennifer Mueller from Rensselaer Polytechnic Institute, NY, USA. The results of the project will be published later.

We present here briefly the contents of this work. In Chapter 2 we give an introduction to Nachman's proof [43]. A more detailed treatment can be found in [45]. Chapter 3 presents new results of practical importance related to Nachman's method:

- estimates, representation formulas and symmetry properties for Faddeev fundamental solutions g_k and G_k ;
- information about the size, smoothness and asymptotic behaviour of the functions $\mu(x, k)$ (important intermediate objects in Nachman's proof);
- observations on how various properties of the conductivity manifest themselves in the non-physical scattering transform $\mathbf{t}(k)$, another function needed in the reconstruction;
- new estimates on $|\mathbf{t}(k)|$ for both large and small $|k|$.

These ideas are used in Chapter 4 for designing numerical algorithms useful for various steps of the reconstruction, namely

- computing numerically values of the Faddeev Green's functions g_k and G_k ,
- computing the functions $\mu(x, k)$ and $\mathbf{t}(k)$ for given test conductivities. One radially symmetric example is computed numerically. The reason for choosing a symmetric example is to avoid intensive computations and utilize the symmetry theoretically; the algorithms work for nonsymmetric cases, too.

2. A. Nachman's reconstruction method

2.1. Definitions

In 1996, A. Nachman published a constructive proof of the inverse conductivity problem in two-dimensional space. The main theorem of his work is

Theorem 2.1 [43, Theorem 1] *Let Ω be a bounded, simply connected Lipschitz domain in $\mathbb{R}^2$. Let γ_1 and γ_2 be in $W^{2,p}(\Omega)$ for some $p > 1$, and have positive lower bounds.*

$$\text{If } \Lambda_{\gamma_1} = \Lambda_{\gamma_2} \text{ then } \gamma_1 = \gamma_2.$$

The proof gives a constructive procedure for recovering γ from the knowledge of Λ_γ .

We will make some simplifying assumptions. The first part of the proof of Nachman's theorem is to find γ and $\partial\gamma/\partial n$ on the boundary of Ω and reduce the problem to the case $\gamma \equiv 1$ in a neighbourhood of $\partial\Omega$; the reduction is done in such a way that we can take $\partial\Omega$ to be smooth. We will thus assume

$$\gamma \equiv 1 \text{ near the } C^\infty \text{ boundary } \partial\Omega.$$

Furthermore, we prefer working with smooth functions instead of the Sobolev ones whenever possible and will therefore consider only smooth conductivities:

$$\gamma \in C^\infty(\overline{\Omega}).$$

From the point of view of applications there is not much difference in considering C^∞ conductivities instead of $W^{2,p}$ ones since in practice the conductivities (inner organs of people) are discontinuous and by Sobolev imbedding theorem the $W^{2,p}$ functions are continuous.

The solution of the inverse problem is based on the use of an intermediate object called the *non-physical scattering transform* $\mathbf{t}$, which is not directly measurable in experiments. See Figure 3 for a schematic illustration. We give the definition of $\mathbf{t}$ and explain the two main steps of the procedure, namely finding $\mathbf{t}$ from the boundary measurements Λ_γ and determining γ from the knowledge of $\mathbf{t}$.

To define $\mathbf{t}$ we consider the Schrödinger equation associated with the inverse conductivity problem. If u is a solution of $\nabla \cdot \gamma \nabla u = 0$ in Ω , defining $\tilde{u} = \gamma^{1/2}u$ yields

$$(-\Delta + q)\tilde{u} = 0 \text{ in } \Omega \tag{2}$$

with the potential $q \in C_0^\infty(\Omega)$ given by

$$q = \frac{\Delta\gamma^{1/2}}{\gamma^{1/2}}. \tag{3}$$

Moreover, our assumption that γ be one near Ω allows us to extend smoothly γ by one and q by zero to the whole plane. The exponentially behaving solutions of (2) introduced by Faddeev [15] are the key to the reconstruction:

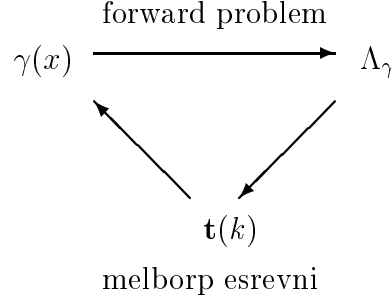


Figure 3: Overall strategy of the proof.

Theorem 2.2 [43, Theorem 1.1] *Let $q \in L^p(\mathbb{R}^2)$, $1 < p < 2$, be such that there exists a real-valued $\gamma \in L^\infty(\mathbb{R}^2)$ with $q = \gamma^{-1/2} \Delta \gamma^{1/2}$, $\gamma(x) \geq c > 0$ and $\nabla \gamma \in L^p(\mathbb{R}^2)$.*

Then for any $k \in \mathbb{C} \setminus 0$ there is a unique solution $\mu(x, k)$ of

$$(-\Delta + q)e^{ikx}\mu(x, k) = 0 \quad (4)$$

with $\mu(\cdot, k) - 1 \in L^{\tilde{p}} \cap L^\infty$. Moreover, $\mu - 1 \in W^{1, \tilde{p}}$ and

$$\|\mu(\cdot, k) - 1\|_{W^{s, \tilde{p}}} \leq c|k|^{s-1}\|q\|_{L^p} \quad (5)$$

for $0 \leq s \leq 1$ and k sufficiently large.

Here $e^{ikx} = \exp(i(k_1 + ik_2)(x_1 + ix_2))$ and $\tilde{p}$ is defined by $1/\tilde{p} = 1/p - 1/2$. The Sobolev space $W^{1, \tilde{p}}$ is defined by

$$W^{1, \tilde{p}}(\mathbb{R}^2) = \{f \in L^{\tilde{p}}(\mathbb{R}^2) \mid \partial_i f \in L^{\tilde{p}}(\mathbb{R}^2), \quad i = 1, 2\},$$

and the space $W^{s, \tilde{p}}$ is the interpolation space between $L^{\tilde{p}}$ and $W^{1, \tilde{p}}$ [4]. Note that the condition $\mu - 1 \in W^{1, \tilde{p}}$ combined with Sobolev imbedding theorem means that $\mu - 1$ is continuous and tends to zero asymptotically when $|k| \rightarrow \infty$. Furthermore, since $\mu \sim 1$, we see that the solutions $\psi(x, k) := e^{ikx}\mu(x, k) \sim e^{ikx}$ have some sort of exponential behaviour. Perhaps the most important content of Theorem 2.2 is the existence of the Faddeev solutions for *all* nonzero k since this was known only for large $|k|$ from [50].

We define the non-physical scattering transform for all $k \neq 0$ by

$$\mathbf{t}(k) := \int_{\mathbb{R}^2} e_k(x)\mu(x, k)q(x)dx, \quad (6)$$

where $e_k(x) = \exp(i(kx + \bar{k}\bar{x}))$. Note that because μ is asymptotically near to one, $\mathbf{t}(k)$ resembles the Fourier transform of $q(x)$ evaluated at the point $(-2k_1, 2k_2) \in \mathbb{R}^2$, for large $|k|$.

We will take a closer look at the construction of the functions μ in section 3.2 and prove that they are smooth in x .

2.2. From Λ_γ to t

First we want to find a relation between the Dirichlet-to-Neumann maps of the original conductivity problem and the Schrödinger problem described above. It turns out that the maps coincide.

Note that zero is not a Dirichlet eigenvalue of the operator $-\Delta + q$, that is, there are no nontrivial solutions $\tilde{u}$ of (2) with vanishing trace. This is evident because for any such $\tilde{u}$ the function $\gamma^{-1/2}\tilde{u}$ would be a nonvanishing solution with trace zero of the conductivity equation, which is a contradiction to the unique solvability of the Dirichlet problem (1). Thus we can define the Dirichlet-to-Neumann map $\Lambda_q : H^{1/2}(\partial\Omega) \rightarrow H^{-1/2}(\partial\Omega)$ for equation (2) by

$$\langle g, \Lambda_q f \rangle = \int_{\Omega} \nabla v \cdot \nabla \tilde{u} + \int_{\Omega} q v \tilde{u}, \quad (7)$$

where $\tilde{u} \in H^1(\Omega)$ is the weak solution of (2) with Dirichlet data $\tilde{u}|_{\partial\Omega} = f$ and v is any $H^1(\Omega)$ function with trace g . Furthermore, by differentiating

$$\frac{\partial \tilde{u}}{\partial \nu} = \frac{1}{2} \gamma^{-1/2} \frac{\partial \gamma}{\partial \nu} u + \gamma^{1/2} \frac{\partial u}{\partial \nu} = \frac{\partial u}{\partial \nu}$$

since $\gamma|_{\partial\Omega} \equiv 1$ and $\partial\gamma/\partial\nu|_{\partial\Omega} \equiv 0$. Thus $\Lambda_q = \Lambda_\gamma$, so our task is to find the scattering transform out of Λ_q .

Define a single-layer operator S_k for $k \in \mathbb{C} \setminus 0$ by

$$(S_k \phi)(x) := \int_{\partial\Omega} G_k(x - y) \phi(y) d\sigma(y), \quad (8)$$

where the function $G_k(x)$ is Faddeev's exponential fundamental solution for $-\Delta$. We will discuss this function in detail in section 3.1.1. Denote the Dirichlet-to-Neumann map of the zero potential (corresponding to homogeneous conductivity 1) in Ω by Λ_0 . We have

Theorem 2.3 [43, Theorem 5] *Let Ω be a bounded Lipschitz domain in $\mathbb{R}^2$ and let $q \in L^p(\Omega)$ for some $p > 1$. Assume 0 is not a Dirichlet eigenvalue of $-\Delta + q$ in Ω and denote $\psi(x, k) = e^{ikx} \mu(x, k)$ with μ given by Theorem 2.2. Then, for any $k \in \mathbb{C} \setminus 0$:*

(i) $S_k(\Lambda_q - \Lambda_0)$ is a compact operator: $H^{1/2}(\partial\Omega) \rightarrow H^{1/2}(\partial\Omega)$;

(ii) The trace on $\partial\Omega$ of the function $\psi(\cdot, k)$ satisfies the integral equation

$$\psi(\cdot, k)|_{\partial\Omega} = e^{ikx} - S_k(\Lambda_q - \Lambda_0)\psi(\cdot, k); \quad (9)$$

(iii) $I + S_k(\Lambda_q - \Lambda_0)$ is invertible on $H^{1/2}(\partial\Omega)$ if and only if the Schrödinger problem (4) has a solution (here q is extended to be zero outside Ω);

(iv) If $\psi(\cdot, k)|_{\partial\Omega}$ has been determined, then $\mathbf{t}(k)$ can be recovered from the formula

$$\mathbf{t}(k) = \int_{\partial\Omega} e^{i\bar{k}x} (\Lambda_q - \Lambda_0) \psi(\cdot, k) d\sigma. \quad (10)$$

Clearly the numerical realization of Theorem 2.3 is an important application for the Faddeev Green's function algorithm developed in section 3.1. Moreover, the possibility to compute $\mathbf{t}(k)$ numerically from known γ presented in section 4.2 is vital in testing any algorithm for Theorem 2.3.

2.3. From $\mathbf{t}$ to γ

The non-physical scattering transform (6) contains enough information to recover γ . Namely, Theorem 2.1 of [43] shows that the $\bar{\partial}$ equation

$$\frac{\partial}{\partial \bar{k}} \mu(x, k) = \frac{1}{4\pi \bar{k}} \mathbf{t}(k) e_{-k}(x) \overline{\mu(x, k)}, \quad k \neq 0, \quad (11)$$

where $e_k(x) = \exp(i(kx + \bar{k}\bar{x}))$, holds (with respect to the variable k) in the weighted Sobolev space

$$W_{-\beta}^{1, \tilde{p}} = \{\langle \cdot \rangle^{-\beta} f \in W^{1, \tilde{p}}\}, \quad \beta > 2/\tilde{p},$$

where $1/\tilde{p} = 1/p - 1/2$ and $\langle \cdot \rangle = (1 + |\cdot|^2)^{1/2}$. By Theorem 4.1 of [43] the solution of (11) is given by

$$\mu(x, k) = 1 - \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} \frac{\mathbf{t}(k')}{(k' - k)\bar{k}'} e_{-x}(k') \overline{\mu(x, k')} dk'_1 dk'_2 \quad (12)$$

for all $x \in \mathbb{R}^2, k \in \mathbb{C} \setminus \{0\}$.

Taking formally k to zero in (4) yields

$$(-\Delta + q)\mu(x, 0) = (-\Delta + \frac{\Delta \gamma^{1/2}}{\gamma^{1/2}})\mu(x, 0) = 0,$$

which suggests that $\lim_{k \rightarrow 0} \mu(x, k) = \mu(x, 0) = \sqrt{\gamma(x)}$ because of the uniqueness in Theorem 2.2 and the fact that $\gamma - 1$ in $L^{\tilde{p}} \cap L^\infty$. Section 3 of [43] is devoted to the proof that this indeed is the case. Thus we can recover the conductivity by taking k to zero in (12):

$$\gamma^{1/2}(x) = \lim_{k \rightarrow 0} \mu(x, k) = 1 - \frac{1}{4\pi^2} \int_{\mathbb{C}} \frac{\mathbf{t}(k)}{|k|^2} e_{-x}(k) \overline{\mu(x, k)} dk. \quad (13)$$

Note that to evaluate the last expression, $\mu(x, k)$ is needed for all values of $k \in \mathbb{C} \setminus 0$.

An application of the numerical methods presented in Chapter 4 is to compute μ and consequently $\mathbf{t}$ for known conductivities. This way a computational realization of this step can be tested.

2.4. Conditional stability

G. Alessandrini showed in [2] an example demonstrating the ill-posedness of the inverse conductivity problem: let Ω be the unit disc in $\mathbb{R}^2$ and define two conductivities $\gamma_0, \gamma_r \in L^\infty(\Omega)$ by $\gamma_0 \equiv 1$ and

$$\gamma_r(x) = \begin{cases} s & \text{for } |x| \leq r, \\ 1 & \text{for } r < |x| \leq 1, \end{cases}$$

where $0 < s < \infty$ and $0 < r < 1$. By explicitly calculating the Dirichlet-to-Neumann map for both conductivities he showed that

$$\|\Lambda_{\gamma_r} - \Lambda_{\gamma_0}\|_{H^{1/2}(\partial\Omega) \rightarrow H^{-1/2}(\partial\Omega)} \leq |1 - s|r \rightarrow 0, \quad r \rightarrow 0,$$

while

$$\|\gamma_0 - \gamma_r\|_{L^\infty(\Omega)} = |1 - s|$$

is independent of r . It follows that there can be no continuous real function with $f(0) = 0$ and an inequality of type

$$\|\gamma_0 - \gamma_r\|_{L^\infty(\Omega)} \leq f(\|\Lambda_{\gamma_r} - \Lambda_{\gamma_0}\|_{H^{1/2}(\partial\Omega) \rightarrow H^{-1/2}(\partial\Omega)}) \quad (14)$$

ensuring that a small error in measured data leads to small L^∞ error in the reconstruction.

From the practical point of view the above seems devastating. However, it is possible to derive stability estimates of the form (14) if there is a priori knowledge about the conductivity. This is in general a reasonable assumption because in medical EIT there is a lot of knowledge about the type of conductivity changes inside the patient.

There is a stability analysis available for the reconstruction method at hand: the work of Nachman and Liu [43, 40] shows that

Theorem 2.4 [40, Theorem 1] *Let γ_1, γ_2 satisfy the a priori assumptions $\|\gamma_j\|_{W^{2,p}(\Omega)} \leq M$ for some fixed $1 < p < 2$ and $\gamma_j(x) \geq 1/M$ in Ω , $j = 1, 2$. If on $\partial\Omega$ we have $\gamma_j = 1$ and $\partial\gamma_j/\partial\nu = 0$, then*

$$\|\gamma_r - \gamma_0\|_{L^\infty(\Omega)} \leq C_M \omega(\|\Lambda_{\gamma_r} - \Lambda_{\gamma_0}\|_{H^{1/2}(\partial\Omega) \rightarrow H^{-1/2}(\partial\Omega)}). \quad (15)$$

The function ω is such that

$$\omega(\varepsilon) \leq |\log \varepsilon|^{-\delta} \quad (16)$$

for every $\delta, 0 < \delta < 2/p'$, and sufficiently small ε ; the constant C_M depends only on M, p and Ω .

It is interesting to note that, roughly speaking, the inversion step from Λ_γ to $\mathbf{t}(k)$ has logarithmic stability and the step from $\mathbf{t}(k)$ to γ has linear stability.

2.5. Practical aspects

Nachman's conductivities are continuous, and so are the ones considered by Brown and Uhlmann [7]. In practice the interesting conductivities (inner organs of a person) are discontinuous, and it not clear how numerical realizations of these reconstruction methods will perform with data corresponding to conductivities with jumps.

In Nachman's approach the boundary data is the Dirichlet-to-Neumann map Λ_γ , which cannot be measured practically in a stable way. Its smoothing inverse operator R_γ , the Neumann-to-Dirichlet map, is the one that has to be used in physical experiments. From this map, Λ_γ can be computed after the measurements since the two are inverse maps to each other.

It is easier to recover details of γ located near the boundary $\partial\Omega$. In fact, as pointed out in [18] and the references therein, the contribution of conductivity details to the boundary measurements gets exponentially smaller as the distance from the boundary increases. This is one reason why exponentially growing solutions are needed in the reconstruction. However, in EIT measurements it is for safety reasons impossible to use arbitrary large electric currents, so the boundary values of the exponentially growing solutions have to be approximated. This process is numerically unstable for large $|k|$.

The computational cost of Nachman's reconstruction will be great. This is a consequence of the four-dimensional nature of the method: both the x - and k -variables range in a 2D grid. It may not be a possible choice for a real-time EIT application, but in any case it provides a reliable initial choice for a faster iterative approach.

3. Theoretical observations

The purpose of this chapter is to derive theoretical results that help the realization of Nachman's reconstruction scheme. Our goal is to answer such practical questions as

- How to compute efficiently and accurately point values of Faddeev Green's functions G_k and g_k ?
- Finding the functions μ from the $\bar{\partial}$ equation involves solving a weakly singular integral equation with $\mathbf{t}(k)$ appearing in the kernel. How singular is the kernel at $k = 0$? How much error does the truncation of the integral introduce? In other words, what is the exact rate of decay of $|\mathbf{t}(k)|$ when $|k|$ grows?
- How much can we deduce qualitative information about γ from symmetries of $\mathbf{t}(k)$?

Section 3.1 is an analysis of the Faddeev Green's functions g_k and G_k . There are more results than needed in this work; we believe that they are valuable for the field of inverse problems in 2D electrical prospection or nonlinear evolution equations of Novikov-Veselov type [58, 53].

In section 3.2 we derive both quantitative and qualitative facts concerning $\mathbf{t}(k)$ and $\mu(x, k)$.

3.1. Faddeev Green's functions

The two fundamental solutions $g_k(x)$ and $G_k(x)$ originally introduced by Faddeev [15] are in the core of Nachman's reconstruction procedure. In this section we define and analyse these functions and use the results later in both analytical and numerical considerations.

In this work we deal with functions $g_k(x)$ and $G_k(x)$ defined on $\mathbb{R}^2$. Similar results on their three-dimensional analogues can be found in [44].

3.1.1 Definitions and basic properties Define the function $g_k(x)$ for all $x \in \mathbb{R}^2$ and $k \in \mathbb{C} \setminus 0$ by the formula

$$g_k(x) = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} \frac{e^{ix \cdot \xi}}{\xi(\bar{\xi} + 2k)} d\xi, \quad (17)$$

where $x \cdot \xi = x_1 \xi_1 + x_2 \xi_2$ in the exponent and in the denominator we have $\xi = \xi_1 + i\xi_2$ and complex multiplication. The function $g_k(x)$ is a fundamental solution for the operator

$$L_k = -\Delta - 4ik\bar{\partial} = -\Delta - 4ik\frac{1}{2}\left(\frac{\partial}{\partial x_1} + i\frac{\partial}{\partial x_2}\right), \quad (18)$$

that is,

$$L_k g_k(x) = \delta(x). \quad (19)$$

This can be seen by noting that the Fourier symbol of L_k is

$$\widehat{L_k}(\xi) = |\xi|^2 + 2k(\xi_1 + i\xi_2) = \xi(\bar{\xi} + 2k)$$

and formula (17) is essentially the inverse Fourier transform of $1/\widehat{L_k}$ understood in the sense of tempered distributions: $g_k(x) = \mathcal{F}^{-1}(\xi(\bar{\xi} + 2k))^{-1}$.

In Nachman's reconstruction scheme the function g_k is a key object. It is used to find the exponentially behaving solutions $\psi(x, k) = e^{ikx} \mu(x, k)$ mentioned in Chapter 2; we will discuss this construction in sections 3.2 and 4.2. Another important application for $g_k(x)$ is going from the boundary measurements Λ_γ to the non-physical scattering transform $\mathbf{t}(k)$; see Theorem 2.3 for details. The function $G_k(x)$ needed for this is defined for all $x \in \mathbb{R}^2$ and $k \in \mathbb{C} \setminus 0$ by

$$G_k(x) := e^{ikx} g_k(x). \quad (20)$$

Note that we used again the complex notation $kx = (k_1 + ik_2)(x_1 + ix_2)$, so that the modulus of the exponential multiplier is not necessarily one. Given this way, G_k will be a fundamental solution for the negative Laplacian:

$$-\Delta G_k(x) = -4\partial\bar{\partial}e^{ikx}g_k(x) = -4\partial e^{ikx}\bar{\partial}g_k(x) = e^{ikx}(-\Delta - 4ik\bar{\partial})g_k(x) = \delta(x), \quad (21)$$

where we used (19) and the fact that $\bar{\partial}e^{ikx} = 0$ since e^{ikx} is complex analytic in $x = x_1 + ix_2$.

To compare the Faddeev fundamental solution with the regular one, let G_0 the standard fundamental solution for $-\Delta$:

$$G_0(x) = -\frac{1}{2\pi} \log|x|,$$

and denote the difference between the two by

$$H_k(x) := G_k(x) - G_0(x), \quad k \in \mathbb{C} \setminus 0. \quad (22)$$

Now H_k is a harmonic distribution since $\Delta H_k = -\delta + \delta = 0$, and by Weyl's lemma we know that $H_k(x)$ is a smooth, harmonic function in the whole plane. This helps us understand the behaviour of G_k and g_k at the origin: both have a logarithmic singularity there. This follows from equation (22) and the identity

$$g_k(x) = e^{-ikx}H_k(x) - \frac{e^{-ikx}}{2\pi} \log|x|, \quad (23)$$

respectively.

For convenience, we collect our observations about H_k together as a lemma.

Lemma 3.1 *Let $x \in \mathbb{R}^2$ and $k \in \mathbb{C} \setminus 0$. Then the function $H_k(x)$ given in $\mathbb{R}^2 \setminus 0$ by (22) is smooth and harmonic at every point $x \in \mathbb{R}^2$.*

In section 3.1.6 we will compute $H_k(0)$.

Change of variables in the defining integral (17) shows that the Faddeev fundamental solution g_k has several symmetries:

Theorem 3.2 *Let $x \in \mathbb{R}^2 \setminus 0$ and $k \in \mathbb{C} \setminus 0$. The Faddeev Green's function g_k given by (17) has the properties*

$$g_k(x) = g_1(kx), \quad (24)$$

$$g_k(x) = \overline{g_{\bar{k}}(-\bar{x})}, \quad (25)$$

$$g_k(x) = e_{-k}(x)\overline{g_k(x)}, \quad (26)$$

where $e_k(x) = \exp(i(kx + \bar{k}\bar{x}))$.

This together with the computation

$$H_k(x) = e^{ikx} g_1(kx) + \frac{1}{2\pi} \log |k| |x| - \frac{1}{2\pi} \log |k| = H_1(kx) - \frac{1}{2\pi} \log |k|$$

shows that we can restrict our attention to the functions g_1, G_1 and H_1 :

$$G_k(x) = G_1(kx), \quad (27)$$

$$H_k(x) = H_1(kx) - \frac{1}{2\pi} \log |k|. \quad (28)$$

The functions $G_k(x)$ and $H_k(x)$ are real-valued for all $k \in \mathbb{C} \setminus 0$. This is seen by computing

$$\overline{G_k(x)} = e^{-i\bar{k}\bar{x}} \overline{g_k(x)} = e^{-i\bar{k}\bar{x}} e_k(x) g_k(x) = e^{ikx} g_k(x) = G_k(x).$$

3.1.2 Derivatives In this section we will derive explicit formulas for the first-order partial derivatives of $g_k(x)$:

Theorem 3.3 *Let $x \in \mathbb{R}^2 \setminus 0, k \in \mathbb{C} \setminus 0$, and define the function $g_k(x)$ by (17). The first order partial derivatives of $g_k(x)$ are given by*

$$\frac{\partial g_k}{\partial x_1}(x) = -\frac{1}{4\pi x} - \frac{e_{-k}(x)}{4\pi \bar{x}} - ik g_k(x), \quad (29)$$

$$\frac{\partial g_k}{\partial x_2}(x) = \frac{1}{4\pi i x} - \frac{e_{-k}(x)}{4\pi i \bar{x}} + k g_k(x), \quad (30)$$

where $e_{-k}(x) = \exp(-i(kx + \bar{k}\bar{x}))$.

Before the proof we need a couple of lemmas.

Lemma 3.4 *We have*

$$\mathcal{F}^{-1} \frac{2}{i\xi} = \frac{1}{\pi x} \quad \text{and} \quad \mathcal{F}^{-1} \frac{2}{i\xi} = \frac{1}{\pi \bar{x}}, \quad (31)$$

where $x = x_1 + ix_2$ and the inverse Fourier transforms are taken in the sense of tempered distributions.

Proof. We will prove the first of the identities in (31). The second one follows from this by taking complex conjugates. Denote

$$h(x) := \mathcal{F}^{-1} \frac{2}{i\xi} = \mathcal{F}^{-1} \frac{2}{i(\xi_1 + i\xi_2)} \quad (32)$$

and note that the properties of the Fourier transform imply that h is a tempered distribution satisfying $\bar{\partial} h = \delta$. On the other hand, it is well-known [25, Chapter III] that $f(x) := 1/\pi x$ is a fundamental solution for the operator $\bar{\partial}$, that is $\bar{\partial} f = \delta$.

Our calculations show that $h(x) = f(x) + u(x)$, where $u = h - f$ satisfies

$$\bar{\partial}u = \bar{\partial}h - \bar{\partial}f = \delta - \delta = 0. \quad (33)$$

By elliptic regularity we know that u is differentiable and so (33) implies that u is an entire function. On the Fourier transform side (33) takes the form

$$\frac{i}{2}(\xi_1 + i\xi_2)\hat{u} = 0, \quad (34)$$

which implies that $\hat{u}$ is supported in the origin, so $u(x)$ is a polynomial of x_1 and x_2 . Lemma 7.1.16 in [25] states that if $u \in \mathcal{S}(\mathbb{R}^2)$ is homogeneous of degree -1 , then the Fourier transform $\hat{u}$ is also homogeneous of degree -1 . This is the case for $f(x)$ and cannot be true for the sum of f and a nonzero polynomial; so $u \equiv 0$ and the proof is complete. **Q.E.D.**

Lemma 3.5 *Let $x \in \mathbb{R}^2 \setminus 0$ and define the function g_1 by (17). We have*

$$\bar{\partial}g_1(x) = -\frac{e^{-i2x_1}}{4\pi\bar{x}}, \quad (35)$$

$$\partial g_1(x) = -\frac{1}{4\pi x} - ig_1(x), \quad (36)$$

$$\Delta g_1(x) = \frac{ie^{-i2x_1}}{\pi\bar{x}}. \quad (37)$$

Proof. We start with determining $\bar{\partial}g_1(x)$. Compute

$$\widehat{\bar{\partial}g_1} = \frac{i\xi/2}{\xi(\bar{\xi} + 2)} = \frac{i}{2(\bar{\xi} + 2)},$$

and use Lemma 3.4 and the translation property of the Fourier transform to get

$$\bar{\partial}g_1(x) = \frac{i}{2}\mathcal{F}^{-1}\left(\frac{1}{\bar{\xi} + 2}\right) = -\frac{e^{-i(2,0)\cdot x}}{4}\mathcal{F}^{-1}\frac{2}{i\xi} = -\frac{e^{-i2x_1}}{4\pi\bar{x}}, \quad (38)$$

which is (35).

The Laplacian of g_1 is now easily computed with aid of (38) and the fact that $\bar{x}^{-1}$ is antiholomorphic at $x \neq 0$:

$$\Delta g_1(x) = 4\partial\bar{\partial}g_1(x) = -4\partial\frac{e^{-i(x+\bar{x})}}{4\pi\bar{x}} = -\frac{e^{-i\bar{x}}}{\pi\bar{x}}\partial e^{-ix} = \frac{ie^{-i2x_1}}{\pi\bar{x}},$$

and identity (37) is proved.

To evaluate $\partial g_1(x)$ at $x \neq 0$, take complex conjugate of equation (19) and apply equation (26) of Theorem 3.2 to get

$$0 = -\overline{\Delta g_1(x)} + 4i\overline{\partial g_1(x)} = \frac{ie^{i2x_1}}{\pi x} - 4e^{i2x_1}g_1(x) + 4ie^{i2x_1}\partial g_1(x).$$

Now we can solve for ∂g_1 and the claim follows. **Q.E.D.**

Proof. (Theorem 3.3) We start with the special case $k = 1$. Then by Lemma 3.5

$$\begin{aligned} \frac{\partial g_1}{\partial x_1}(x) &= (\partial + \bar{\partial})g_1(x) = -\frac{1}{4\pi x} - \frac{e^{-i2x_1}}{4\pi \bar{x}} - ig_1(x), \\ \frac{\partial g_1}{\partial x_2}(x) &= i(\partial - \bar{\partial})g_1(x) = \frac{1}{4\pi ix} - \frac{e^{-i2x_1}}{4\pi i\bar{x}} + g_1(x). \end{aligned}$$

Now the claim follows from $g_k(x) = g_1(kx)$ and the chain rule. **Q.E.D.**

3.1.3 Representation formulas We use residue calculus to find formulas similar to [5, formula (3.10)].

Theorem 3.6 *Let $x = (x_1, x_2) \in \mathbb{R}^2 \setminus 0$ with $x_1 > 0$. Then $g_1(x)$ is given by*

$$g_1(x) = \frac{e^{-ix_1}}{2\pi} \operatorname{Re} \left[e^{ix_1} \int_0^\infty \frac{e^{-t(x_1+ix_2)}}{t-i} dt \right]. \quad (39)$$

Formula (39) combined with a Laplace transform identity gives another representation of g_k :

Corollary 3.7 *Let $x \in \mathbb{R}^2 \setminus 0, x_1 > 0$. Then $g_1(x)$ is given by*

$$g_1(x) = \frac{e^{-ix_1}}{2\pi} \int_{x_1}^\infty \frac{s \cos s + x_2 \sin s}{s^2 + x_2^2} ds. \quad (40)$$

Proof. (Theorem 3.6) Write formally

$$\begin{aligned} (2\pi)^2 g_1(x) &= \int_{-\infty}^\infty \int_{-\infty}^\infty \frac{e^{i(x_1\xi_1+x_2\xi_2)}}{(\xi_1+1)^2 + (\xi_2+i)^2} d\xi_1 d\xi_2 \\ &= e^{-ix_1} \int_{-\infty}^\infty e^{ix_2\xi_2} \left[\int_{-\infty}^\infty \frac{e^{ix_1 t}}{t^2 + (\xi_2+i)^2} dt \right] d\xi_2. \end{aligned}$$

To evaluate the inner integral note that

$$f(z) = \frac{e^{ix_1 z}}{z^2 + (\xi_2+i)^2} = \frac{e^{ix_1 z}}{(z-z_0)(z+z_0)},$$

where $z_0 = -1 + i\xi_2$, is a meromorphic complex function. Define for all $R > 0$ the path $\Gamma(R) = [-R, R] \cup \{Re^{i\varphi} : 0 \leq \varphi \leq \pi\}$ with positive orientation. Observe that

$x_1 > 0$ implies the exponential decay of $f(z)$ in the domain $\text{Im}(z) > 0$. Compute for $\xi_2 > 0$ and large R

$$2\pi i \text{Res}_{z=z_0} f(z) = \int_{\Gamma(R)} f(z) dz = \int_{-R}^R \frac{e^{ix_1 t}}{t^2 + (\xi_2 + i)^2} dt + \mathcal{O}(1/R).$$

Taking the limit $R \rightarrow \infty$ we see that

$$\int_{-\infty}^{\infty} \frac{e^{ix_1 t}}{t^2 + (\xi_2 + i)^2} dt = 2\pi i \frac{e^{ix_1(-1+i\xi_2)}}{(-2+2i\xi_2)} = \frac{\pi e^{-ix_1-x_1\xi_2}}{\xi_2 + i}, \quad \xi_2 > 0.$$

In a similar fashion we obtain

$$\int_{-\infty}^{\infty} \frac{e^{ix_1 t}}{t^2 + (\xi_2 + i)^2} dt = -\frac{\pi e^{ix_1+x_1\xi_2}}{\xi_2 + i}, \quad \xi_2 < 0.$$

Thus for $x_1 > 0$

$$\begin{aligned} (2\pi)^2 g_1(x) &= \pi e^{-ix_1} \left(-\int_{-\infty}^0 \frac{e^{ix_2\xi_2+ix_1+x_1\xi_2}}{\xi_2 + i} d\xi_2 + \int_0^{\infty} \frac{e^{ix_2\xi_2-ix_1-x_1\xi_2}}{\xi_2 + i} d\xi_2 \right) \\ &= \pi e^{-ix_1} \left(e^{ix_1} \int_0^{\infty} \frac{e^{-ix_2\xi_2-x_1\xi_2}}{\xi_2 - i} d\xi_2 + e^{-ix_1} \int_0^{\infty} \frac{e^{ix_2\xi_2-x_1\xi_2}}{\xi_2 + i} d\xi_2 \right) \\ &= 2\pi e^{-ix_1} \text{Re} \left[e^{ix_1} \int_0^{\infty} \frac{e^{-ix_2\xi_2-x_1\xi_2}}{\xi_2 - i} d\xi_2 \right], \end{aligned}$$

and the claim follows. **Q.E.D.**

Proof. (Corollary 3.7) Notice that the Laplace transform of the constant function 1 is $1/z$ for $\text{Re}(z) > 0$, i.e.

$$\frac{1}{z} = \int_0^{\infty} e^{-zw} dw, \quad \text{Re}(z) > 0.$$

Using this we can compute

$$\int_0^{\infty} \frac{e^{-xt}}{t-i} dt = \int_0^{\infty} e^{iw} \int_0^{\infty} e^{-t(x_1+ix_2+w)} dt dw = \int_0^{\infty} \frac{e^{iw}}{x_1+ix_2+w} dw.$$

Substitution to formula (39) and a change of coordinates $s = x_1 + w$ yield

$$\begin{aligned} e^{ix_1} 2\pi g_1(x) &= \text{Re} \left[\int_0^{\infty} \frac{e^{i(x_1+w)}}{ix_2 + (x_1+w)} dw \right] = \text{Re} \left[\int_{x_1}^{\infty} \frac{e^{is}}{ix_2 + s} ds \right] = \\ &= \int_{x_1}^{\infty} \frac{\text{Re}[(s-ix_2)e^{is}]}{s^2 + x_2^2} ds = \frac{e^{-ix_1}}{2\pi} \int_{x_1}^{\infty} \frac{s \cos s + x_2 \sin s}{s^2 + x_2^2} ds. \end{aligned}$$

Q.E.D.

3.1.4 Estimates We start with a simple estimate for $|g_k(x)|$ which is a straightforward consequence of Theorem 3.6.

Theorem 3.8 *Let $x \in \mathbb{R}^2 \setminus 0$ and $k \in \mathbb{C} \setminus 0$. Then*

$$|g_k(x)| \leq \frac{1}{\pi|k||x|}. \quad (41)$$

Combining this result with Theorem 3.3 yields immediately

Corollary 3.9 *Let $k \in \mathbb{C} \setminus 0$. For $x \in \mathbb{R}^2$ satisfying $|x| \geq 1$ and an arbitrary multi-index α we have*

$$|\partial^\alpha g_k(x)| \leq \frac{C(\alpha, k)}{|x|}. \quad (42)$$

The integral formula (17) is formal because the integrand belongs neither to $L^1(\mathbb{R}^2)$ nor $L^2(\mathbb{R}^2)$. Using the Riesz-Thorin interpolation theorem, however, the inverse Fourier transform can be understood as an interpolated operator from $L^{\lambda'}$ to L^λ for $1 < \lambda' < 2$, $2 < \lambda < \infty$ and $1/\lambda' + 1/\lambda = 1$. Since the integrand in (17) belongs to $L^{\lambda'}$, we see that $g_k \in L^\lambda$:

Theorem 3.10 *Let $k \in \mathbb{C} \setminus 0$ and $2 < \lambda < \infty$. The Faddeev Green's function given by (17) satisfies the estimate*

$$\|g_k\|_{L^\lambda} \leq \frac{|k|^{-2/\lambda}}{2\pi} \left(\frac{2\pi}{\lambda' - 1} + \frac{10\pi}{2 - \lambda'} \right)^{1/\lambda'}, \quad (43)$$

where $1/\lambda + 1/\lambda' = 1$.

We now turn to the proofs.

Proof. (Theorem 3.8) Because $g_k(x) = g_1(kx)$ for any $k \in \mathbb{C} \setminus 0$, it is enough to prove

$$|g_1(x)| \leq \frac{1}{\pi|x|}, \quad x \in \mathbb{R}^2 \setminus 0. \quad (44)$$

Let $x_1 > 0$. From formula (39) we see at once that

$$|g_1(x)| \leq \frac{1}{2\pi} \int_0^\infty \frac{e^{-x_1 t}}{|t - i|} dt \leq \frac{1}{2\pi} \int_0^\infty e^{-x_1 t} dt = \frac{1}{2\pi x_1}.$$

To prove a similar estimate in terms of $|x_2|$ we divide the consideration to the cases $x_2 > 0$ and $x_2 < 0$.

Let $x_2 > 0$. The complex function $f(z) = e^{-ix_2 z - x_1 z}(z - i)^{-1}$ is analytic for $z \in \mathbb{C} \setminus i$ and decays exponentially in the quadrant $\operatorname{Re}(z) > 0, \operatorname{Im}(z) < 0$. Define for any $R > 0$ the path

$$\Gamma(R) = [0, -iR] \cup \{Re^{i\varphi} : -\pi/2 \leq \varphi \leq 0\} \cup [R, 0].$$

Then we have

$$\int_0^\infty \frac{e^{-ix_2t-x_1t}}{t-i} dt = \int_0^\infty \frac{e^{-x_2s+ix_1s}}{s+1} ds.$$

This yields as before

$$|g_1(x)| \leq \frac{1}{2\pi} \int_0^\infty e^{-x_2s} ds = \frac{1}{2\pi x_2}, \quad x_2 > 0. \quad (45)$$

Let $x_2 < 0$. In this case $f(z)$ decays exponentially in the quadrant $\operatorname{Re}(z) > 0, \operatorname{Im}(z) > 0$. Define for any $R > 0$ the path

$$\Gamma(R) = [0, R] \cup \{Re^{i\varphi} : 0 \leq \varphi \leq \pi/4\} \cup e^{i\pi/4}[R, 0].$$

Then we have

$$\int_0^\infty \frac{e^{-ix_2t-x_1t}}{t-i} dt = (1+i) \int_0^\infty \frac{e^{-is(x_2+x_1)+s(x_2-x_1)}}{s+is-i} ds,$$

and so, since $|s+is-i| \geq 1/\sqrt{2}$ for all $s > 0$,

$$|g_1(x)| \leq \frac{1}{2\pi} \int_0^\infty \frac{e^{s(x_2-x_1)}}{|s+is-i|} ds \leq \frac{\sqrt{2}}{2\pi} \int_0^\infty e^{s(x_2-x_1)} ds \leq \frac{1}{\sqrt{2}\pi|x_2|}.$$

Now for points x satisfying $x_1 > 0$ and $|x_2| \leq x_1$ we have

$$|g_1(x)| \leq \frac{1}{2\pi x_1} \leq \frac{\sqrt{2}}{2\pi|x|} \leq \frac{1}{\pi|x|}$$

since $|x|/\sqrt{2} \leq x_1$ in this domain. For points satisfying $x_1 > 0$ and $|x_2| > x_1$ we have by a similar argument

$$|g_1(x)| \leq \frac{1}{\sqrt{2}\pi|x_2|} \leq \frac{1}{\pi|x|}.$$

We have thus shown that the desired estimate holds for all (x_1, x_2) satisfying $x_1 > 0$. By continuity of g_1 outside the origin and the particular case $g_1(-x_1, x_2) = \overline{g_1(x_1, x_2)}$ of formula (25) we see that the estimate (44) is valid at all $x \neq 0$. **Q.E.D.**

Proof. (Theorem 3.10) Note that $\|g_k\|_{L^\lambda} = \|g_{|k|}\|_{L^\lambda}$. Furthermore, by (24) we have

$$\|g_{|k|}\|_{L^\lambda}^\lambda = \int_{\mathbb{R}^2} |g_1(|k|x)|^\lambda dx = |k|^{-2} \int_{\mathbb{R}^2} |g_1(x)|^\lambda dx = |k|^{-2} \|g_1\|_{L^\lambda}^\lambda$$

and thus it is enough to prove the claim for g_1 .

By the Riesz-Thorin interpolation theorem we have a continuous operator

$$\mathcal{F}^{-1} : L^{\lambda'} \rightarrow L^\lambda, \quad \|\mathcal{F}^{-1}\|_{L^{\lambda'} \rightarrow L^\lambda} \leq 1/(2\pi). \quad (46)$$

Since $2 < \lambda < \infty$ we have $1 < \lambda' < 2$. We will find a bound for the $L^{\lambda'}$ norm of $\hat{g}_1$ as follows:

$$\|\hat{g}_1\|_{L^{\lambda'}}^{\lambda'} = \left(\int_{|\xi|>4} + \int_{|\xi|\leq 4} \right) \frac{d\xi}{|\xi|^{\lambda'}|\xi+2|^{\lambda'}}. \quad (47)$$

To estimate the first integral in (47) note that in the domain of integration we have $|\xi+2| \geq (|\xi|-2) \geq (|\xi|-|\xi|/2) = |\xi|/2$. Thus

$$\int_{|\xi|>4} \frac{d\xi}{|\xi|^{\lambda'}|\xi+2|^{\lambda'}} \leq 2^{\lambda'} 2\pi \int_4^\infty r^{1-2\lambda'} dr < \frac{2\pi}{\lambda'-1}. \quad (48)$$

For the second integral in (47) compute

$$\begin{aligned} \int_{|\xi|\leq 4} \frac{1}{|\xi|^{\lambda'}} \frac{1}{|\xi+2|^{\lambda'}} d\xi &\leq \int_{|\xi|\leq 4, |\xi+2|>1} \frac{d\xi}{|\xi|^{\lambda'}} + \int_{|\xi+2|\leq 1} \frac{d\xi}{|\xi+2|^{\lambda'}} \leq \\ &\leq 2\pi \left(\int_0^4 r^{1-\lambda'} dr + \int_0^1 r^{1-\lambda'} dr \right) = 2\pi \left(\frac{4^{2-\lambda'}}{2-\lambda'} + \frac{1}{2-\lambda'} \right) < \frac{10\pi}{2-\lambda'}. \end{aligned} \quad (49)$$

Combining (48) and (49) with (46) yields the claim. **Q.E.D.**

3.1.5 An asymptotic expansion Integrating by parts in formula (39) yields for g_1 an asymptotic expansion with error estimate. This is especially suitable for numerical purposes; indeed, we will take Theorem 3.11, below, as the starting point for the numerical algorithm for $g_k(x)$ developed in section 4.1.1.

Theorem 3.11 *Let $x = x_1 + ix_2 \in \mathbb{R}^2 \setminus 0, x_1 > 0$. For any integer $N \geq 0$ we have*

$$g_1(x) = \frac{e^{-ix_1}}{2\pi} \operatorname{Re} \left[-e^{ix_1} \sum_{j=0}^N \frac{j!}{(ix)^{j+1}} \right] + E_N(x), \quad (50)$$

where the remainder $E_N(x)$ is given by

$$E_N(x) = \frac{e^{-ix_1}}{2\pi} \operatorname{Re} \left[\frac{(N+1)!e^{ix_1}}{(-x)^{N+1}} \int_0^\infty \frac{e^{-tx}}{(t-i)^{N+2}} dt \right] \quad (51)$$

and satisfies the estimate

$$|E_N(x)| \leq \frac{(N+1)!2^{(N+1)/2}}{\pi|x|^{N+2}}. \quad (52)$$

Since $g_1(-x_1, x_2) = \overline{g_1(x_1, x_2)}$, similar formulas hold for $x_1 < 0$.

Proof. We claim that for any $N \geq 0$

$$\int_0^\infty \frac{e^{-xt}}{t-i} dt = -\sum_{j=0}^N \frac{j!}{(ix)^{j+1}} + \frac{(N+1)!}{(-x)^{N+1}} \int_0^\infty \frac{e^{-xt}}{(t-i)^{N+2}} dt. \quad (53)$$

We give an induction proof keeping carefully track of the signs. First we consider the case $N = 0$:

$$\int_0^\infty \frac{e^{-xt}}{t-i} dt = \left[\frac{e^{-xt}}{(-x)(t-i)} \right]_0^\infty - \int_0^\infty \frac{-e^{-xt}}{(-x)(t-i)^2} dt = -\frac{1}{ix} - \frac{1}{x} \int_0^\infty \frac{e^{-xt}}{(t-i)^2} dt.$$

Now we make the induction assumption: let the claim hold for all integers smaller or equal to $M \geq 0$. Then the claim for $M+1$ follows from

$$\begin{aligned} \int_0^\infty \frac{e^{-xt}}{t-i} dt &= -\sum_{j=0}^M \frac{j!}{(ix)^{j+1}} + \frac{(M+1)!}{(-x)^{M+1}} \int_0^\infty \frac{e^{-xt}}{(t-i)^{M+2}} dt = \\ &= -\sum_{j=0}^M \frac{j!}{(ix)^{j+1}} + \frac{(M+1)!}{(-x)^{M+1}} \left[\frac{e^{-xt}}{(-x)(t-i)^{M+2}} \right]_0^\infty - \frac{(M+1)!}{(-x)^{M+1}} \int_0^\infty \frac{-(M+2)e^{-xt}}{(-x)(t-i)^{M+3}} dt = \\ &= -\sum_{j=0}^{M+1} \frac{j!}{(ix)^{j+1}} + \frac{(M+2)!}{(-x)^{M+2}} \int_0^\infty \frac{e^{-xt}}{(t-i)^{M+3}} dt. \end{aligned}$$

We have thus established formulas (50) and (51).

The error estimate follows similarly to the proof of Theorem 3.8. The choices of integration paths in the complex plane are the same, but this time we need the estimate $|s + is - i|^{N+2} \geq (1/\sqrt{2})^{N+2}$ for all $s > 0$. **Q.E.D.**

3.1.6 Determining $H_k(0)$

Lemma 3.12 *Let $k \in \mathbb{C} \setminus 0$. Then the value of the function $H_k(x)$ at the origin is*

$$H_k(0) = -\frac{\gamma_0}{2\pi} - \frac{1}{2\pi} \log |k|, \quad (54)$$

where $\gamma_0 \approx 0.577215665$ is Euler's constant.

Proof. Lemma 3.4 in [43] states that for any ε , $0 < \varepsilon < 1$, there is a constant C_ε such that for all $0 < |k| \leq 1/2$ there is an inequality

$$|g_k(x) + (\log |k| + \gamma_0)/2\pi - G_0(x)| \leq C_\varepsilon |k|^\varepsilon \langle x \rangle^\varepsilon, \quad (55)$$

where $\langle x \rangle = (1 + |x|^2)^{1/2}$. Fix now any $x \in \mathbb{R}^2 \setminus 0$ and let $k \in \mathbb{C} \setminus 0$ be arbitrary. Then formula (24) gives

$$H_1(kx) = e^{ikx} g_k(x) + \frac{1}{2\pi} \log |k| |x| =$$

$$= e^{ikx} (g_k(x) + (\log |k| + \gamma_0)/2\pi + \frac{1}{2\pi} \log |x|) + \frac{1}{2\pi} (1 - e^{ikx}) \log |k||x| - \frac{\gamma_0 e^{ikx}}{2\pi}. \quad (56)$$

Choose now any sequence $\{k_j\}_{j=0}^\infty$ of nonzero complex numbers converging to zero. Note that $e^{ikx} - 1 = \mathcal{O}(|kx|)$ for $kx \rightarrow 0$ and that $\lim_{r \rightarrow 0^+} r \log r = 0$, so

$$\lim_{j \rightarrow \infty} (1 - e^{ik_j x}) \log |k_j||x| = 0.$$

Then estimate (55) together with (56) imply

$$H_1(0) = \lim_{j \rightarrow \infty} H_1(k_j x) = -\frac{\gamma_0}{2\pi},$$

and the claim is proved for H_1 . The general result follows from (28) combined with the continuity of $H_k(x)$ with respect to x . **Q.E.D.**

3.2. The scattering transform

Recall from Chapter 2 the definition

$$\mathbf{t}(k) := \int_{\mathbb{R}^2} e_k(x) \mu(x, k) q(x) dx, \quad k \in \mathbb{C} \setminus 0, \quad (57)$$

of the non-physical scattering transform. Here the functions μ are the unique solutions of $(-\Delta + q)e^{ikx}\mu(x, k) = 0$, $x \in \mathbb{R}^2$, satisfying $\mu(\cdot, k) - 1 \in W^{1,\bar{p}}$ (see Theorem 2.2). The function $\mathbf{t}(k)$ is the key intermediate object of the inversion procedure.

We want to understand how the non-physical scattering transform represents the corresponding conductivity. It turns out that $\mathbf{t}(k)$ resembles closely the Fourier transform of $q = \gamma^{-1/2} \Delta \gamma^{1/2}$ evaluated at $(-2k_1, 2k_2) \in \mathbb{R}^2$. Roughly speaking, we can expect this since the exponential in (57) can be written in the form $\exp(-i(-2k_1, 2k_2) \cdot x)$ and the functions $\mu(\cdot, k)$ are close to 1 asymptotically. That $\mathbf{t}(k)$ actually differs from the Fourier transform of q is evident from the fact that the map $q \mapsto \mathbf{t}(k)$ is nonlinear and $\mu \neq 1$ if $q \neq 0$.

The results of this section are aimed to make these speculations more precise and prove such properties of $\mathbf{t}(k)$ that are important from the point of view of numerical work.

The proof of Theorem 4 in [43] implies that $\mathbf{t}(k)$ is continuous outside the origin. We need additional information about its behaviour for both large and small $|k|$ for solving the $\bar{\partial}$ equation (12) numerically. In Section 3.2.1 we present quantitative information about the difference between $\mathbf{t}(k)$ and $\hat{q}$ which leads to an estimate of type $|\mathbf{t}(k)| \leq C|k|^{-m}$ for large $|k|$. This is a consequence of the smoothness of γ . In section 3.2.2 we sharpen results of [43] to get the estimate $|\mathbf{t}(k)| \leq C|k|^{1+\varepsilon}$ for any $0 < \varepsilon < 1$ and $|k|$ small. Thus when solving (12) numerically we don't have to deal with singularities at the origin; also writing formula (13) in polar coordinates yields an integrand vanishing at $|k| = 0$.

In Section 3.2.3 we see that symmetries of γ manifest themselves in the shape of $\mathbf{t}(k)$ almost as in the Fourier transform of q :

- if γ is rotationally symmetric, so is $\mathbf{t}(k)$;
- if γ is preserved in a reflection, roughly the same is true for $\mathbf{t}(k)$.

Two further properties of Fourier transform carry over to $\mathbf{t}(k)$:

- for any $\lambda > 0$ the non-physical scattering transform corresponding to the scaled conductivity $\gamma(\lambda x)$ is $\mathbf{t}(k/\lambda)$;
- translating γ results in multiplication of $\mathbf{t}(k)$ by an exponential function of modulus one.

For a more precise statement see Theorem 3.19. These results are valuable in numerical work for several reasons: given γ we have a priori knowledge about properties of $\mathbf{t}(k)$ and vice versa. This is important for both the forward and the inverse problem. On the other hand, it is possible to produce several examples of pairs $(\gamma, \mathbf{t})$ out of one initial example; this is useful for testing inversion algorithms.

3.2.1 Large $|k|$ estimates

Theorem 3.13 *Let $\Omega \subset \mathbb{R}^2$ be a bounded, simply connected C^∞ domain. Let $\gamma \in C^{m+2}(\mathbb{R}^2)$, $m \geq 1$, have a positive lower bound and satisfy $\gamma - 1 \in C_0^{m+2}(\Omega)$. Define $q = \gamma^{-1/2} \Delta \gamma^{1/2}$ and let $\mathbf{t}(k)$ be given by (57).*

Then there is such a constant $C = C(m, |\Omega|, \|q\|_{C_0^m})$ that

$$|\mathbf{t}(k)| \leq C|k|^{-m} \quad \text{for large } |k|. \quad (58)$$

The rest of this section is devoted to the proof of Theorem 3.13.

Let us recall the construction of the functions μ appearing in (57). For any $k \in \mathbb{C} \setminus 0$ and $1 < p < 2$ we want to find a solution of

$$(-\Delta - 4ik\bar{\partial} + q)\mu(x, k) = 0, \quad x \in \mathbb{R}^2, \quad (59)$$

satisfying $\mu(\cdot, k) - 1 \in W^{1, \tilde{p}}$, where $1/\tilde{p} = 1/p - 1/2$. This is done in the spirit of classical scattering theory: we think of $\mu = \mu_i + \mu_s$ as a sum of an “incoming wave” $\mu_i \equiv 1$ and a “scattered wave” satisfying $\mu_s \in W^{1, \tilde{p}}$, an analogue of the Sommerfeld radiation condition. We will find μ as the solution of the Lippmann-Schwinger type equation

$$\mu - 1 = \mu_s = -g_k * (q\mu). \quad (60)$$

Applying $-\Delta - 4ik\bar{\partial}$ to (60) and recalling (19) shows that a solution of (60) solves (59), too.

To solve (60), note that the operator

$$I + g_k * (q \cdot) : W^{1, \tilde{p}} \rightarrow W^{1, \tilde{p}}$$

is Fredholm of index zero and injective [43, Theorem 1.1 and Lemma 1.5], thus invertible for every $k \in \mathbb{C} \setminus 0$. Furthermore, by Lemma 1.3 of [43] the function $g_k * q \in W^{1,\tilde{p}}$ and we can define

$$\mu(\cdot, k) = 1 - [I + g_k * (q \cdot)]^{-1}(g_k * q). \quad (61)$$

Applying now $I + g_k * (q \cdot)$ to both sides of (61) yields (60).

Since our potential is infinitely smooth, we are able to prove stronger estimates with a simpler proof. For this we need a bound for the norm of the operator $I + g_k * (q \cdot)$ acting on the Sobolev spaces $W^{m,\tilde{p}}$ for $m \geq 1$.

Definition 3.14 Let $D \subset \mathbb{R}^2$ be a domain, $m \geq 0$ an integer and $1 \leq r < \infty$.

The Sobolev space $W^{m,r}(D)$ consists of distributions having all partial derivatives of order less or equal to m in $L^r(D)$. We will use the norm

$$\|u\|_{W^{m,r}(D)} = \sum_{|\alpha| \leq m} \|\partial^\alpha u\|_{L^r(D)}. \quad (62)$$

The Banach space $L(W^{m,r}(D))$ is normed with the operator norm induced by (62).

Theorem 3.15 Let $\Omega \subset \mathbb{R}^2$ be a bounded, simply connected C^∞ domain. Let $\gamma \in C^\infty(\mathbb{R}^2)$ have a positive lower bound and satisfy $\gamma - 1 \in C_0^\infty(\Omega)$. Define $q = \gamma^{-1/2} \Delta \gamma^{1/2}$ and Faddeev Green's function g_k by (17), and let $1 < p < 2$. Fix a $k \in \mathbb{C} \setminus 0$ and let $\mu(x, k)$ be the unique solution of

$$(-\Delta - 4ik\bar{\partial} + q)\mu(x, k) = 0, \quad x \in \mathbb{R}^2, \quad \mu - 1 \in W^{1,\tilde{p}}.$$

Then $\mu - 1 \in W^{m,\tilde{p}}$ for all $m \geq 1$. Furthermore, μ satisfies

$$\|\mu(\cdot, k) - 1\|_{W^{m,\tilde{p}}} \leq 2|k|^{-2/\tilde{p}} \|q\|_{W^{m,1}} \quad \text{for large } |k|. \quad (63)$$

Proof. We start by checking the invertibility of the operator

$$I + g_k * (q \cdot) : W^{m,\tilde{p}} \rightarrow W^{m,\tilde{p}}. \quad (64)$$

By [43, Theorem 1.1] we know that this holds for $m = 1$, and in particular we have injectivity $W^{m,\tilde{p}} \rightarrow W^{1,\tilde{p}}$. The inequalities of Young and Hölder imply for any multi-indices β, β' with $|\beta| + |\beta'| = j \leq m$

$$\begin{aligned} \|g_k * ((\partial^\beta q)(\partial^{\beta'} f))\|_{L^{\tilde{p}}} &\leq \|g_k\|_{L^{\tilde{p}}} \|(\partial^\beta q)(\partial^{\beta'} f)\|_{L^1} \leq \|g_k\|_{L^{\tilde{p}}} \|\partial^\beta q\|_{L^{\tilde{p}'}} \|\partial^{\beta'} f\|_{L^{\tilde{p}}} \\ &\leq \|g_k\|_{L^{\tilde{p}}} \|q\|_{W^{m,\tilde{p}'}} \|f\|_{W^{m,\tilde{p}}}, \end{aligned} \quad (65)$$

so Theorem 3.10 shows that the restriction of $I + g_k * (q \cdot)$ to $W^{m,\tilde{p}}$ has $W^{m,\tilde{p}}$ as target space. Thus it is enough to prove that $g_k * (q \cdot)$ is compact and invertibility of (64) follows by a Fredholm argument. Compactness is evident from the decomposition

$$W^{m,\tilde{p}} \xrightarrow{\iota} C^{m-1} \xrightarrow{q \cdot} W^{m-1,p} \xrightarrow{g_k *} W^{m,\tilde{p}},$$

where we used the Sobolev imbedding theorem and Lemma 1.3 of [43].

To prove (63) we estimate first the operator norm of $[I + g_k * (q \cdot)]^{-1}$. Applying Theorem 3.10 to (65) gives $\|g_k * (q \cdot)\|_{W^{m,\bar{p}} \rightarrow W^{m,\bar{p}}} \leq C \|g_k\|_{L^{\bar{p}}} \|q\|_{W^{m,\bar{p}'}} \leq C |k|^{-2/\bar{p}} \|q\|_{W^{m,\bar{p}'}}$. Thus

$$\|[I + g_k * (q \cdot)]^{-1}\|_{W^{m,\bar{p}} \rightarrow W^{m,\bar{p}}} \leq 2 \quad \text{for large } |k|. \quad (66)$$

An application of Young's inequality and Theorem 3.10 once more gives

$$\|g_k * q\|_{W^{m,\bar{p}}} = \sum_{|\alpha| \leq m} \|g_k * (\partial^\alpha q)\|_{L^{\bar{p}}} \leq \|g_k\|_{L^{\bar{p}}} \sum_{|\alpha| \leq m} \|\partial^\alpha q\|_{L^1} = C |k|^{-2/\bar{p}} \|q\|_{W^{m,1}}. \quad (67)$$

Substituting (66) and (67) to (61) yields the claim. **Q.E.D.**

Proof. (Theorem 3.13) Let $m \geq 1$ be arbitrary. It is enough to derive for large $|k| = |k_1 + ik_2|$ an estimate of type

$$|k_j^m \mathbf{t}(k)| \leq C' = C'(m, |\Omega|, \|q\|_{C_0^m}), \quad j = 1, 2. \quad (68)$$

Integration by parts gives

$$k_1 \int_{\mathbb{R}^2} e_k(x) \psi(x) dx = \frac{i}{2} \int_{\mathbb{R}^2} e_k(x) \frac{\partial \psi}{\partial x_1}(x) dx, \quad \psi \in C_0^1(\mathbb{R}^2),$$

and for k_2 the same with a minus sign. Write $q\mu = q(\mu - 1) + q$ to get

$$|k_j^m \mathbf{t}(k)| \leq 2^{-m} \int_{\mathbb{R}^2} \left| \frac{\partial^m q(x)}{\partial x_j^m} \right| dx + 2^{-m} \int_{\mathbb{R}^2} \left| \frac{\partial^m}{\partial x_j^m} ((\mu(x, k) - 1)q(x)) \right| dx. \quad (69)$$

We can bound the first term by $2^{-m} |\Omega| \sup_{x \in \Omega} \frac{\partial^m q(x)}{\partial x_j^m}$. An application of Theorem 3.15 shows that for large enough $|k|$ the integral in the second term of (69) is bounded by

$$\begin{aligned} & \sum_{\nu=0}^m \binom{\nu}{m} \left\| \frac{\partial^\nu}{\partial x_j^\nu} (\mu(x, k) - 1) \right\|_{L^{\bar{p}}} \left\| \frac{\partial^{m-\nu}}{\partial x_j^{m-\nu}} q(x) \right\|_{L^{\bar{p}'}} \\ & \leq C \|\mu(\cdot, k) - 1\|_{W^{m,\bar{p}}} \|q\|_{W^{m,\bar{p}'}} \leq C \|q\|_{W^{m,1}} \|q\|_{W^{m,\bar{p}'}}. \end{aligned} \quad (70)$$

Estimating the absolute values of derivatives of q by supremums and fixing a p satisfying $1 < p < 2$ in (70) yields the claim. **Q.E.D.**

A digression. We remark that it is possible to follow a path nearer to Nachman's original proof. Namely, we can formally write using the resolvent identity

$$g_k * f = (-\Delta - 4ik\bar{\partial})^{-1} f = -\frac{1}{4}(\partial + ik)^{-1} \bar{\partial}^{-1} f = -\frac{1}{4ik}(I - (\partial + ik)^{-1} \partial) \bar{\partial}^{-1} f.$$

Suppose we had an explicit estimate of type

$$\|\partial^{-1} f\|_{L^{\bar{p}}} \leq A_p \|f\|_{L^p} \quad (71)$$

at our disposal. Then we would get the estimate

$$\|g_k * f\|_{L^{\tilde{p}}} \leq \frac{1}{4|k|} (A_p \|f\|_{L^p} + A_p \|\partial \bar{\partial}^{-1} f\|_{L^p}), \quad (72)$$

since (71) is naturally valid for $\bar{\partial}^{-1}$ as well and, on the other hand, $(\partial + ik)^{-1} = e_{-k} \bar{\partial}^{-1} e_k$ with $e_k(x) = \exp(i(kx + \bar{k}\bar{x}))$. See formulas (1.13) and (1.14) of [43].

Furthermore, we can estimate $\|\partial \bar{\partial}^{-1} f\|_{L^p}$ using the planar Riesz transforms $R_j f, j = 1, 2$, given on the Fourier transform side by

$$\widehat{R_j f}(\xi) = \frac{i\xi_j}{|\xi|} \hat{f}(\xi), \quad \xi \in \mathbb{R}^2, \quad (73)$$

since $\partial \bar{\partial}^{-1} = -(R_1 - iR_2)^2$. An estimate of type

$$\|\partial \bar{\partial}^{-1} f\|_{L^p} \leq B_p \|f\|_{L^p} \quad (74)$$

would lead us to

$$\|g_k * f\|_{L^{\tilde{p}}} \leq \frac{A_p(1 + B_p)}{4|k|} \|f\|_{L^p}, \quad (75)$$

which in turn could be used to prove Theorem 3.15 with 1 instead of $2/\tilde{p}$.

For completeness, we mention how to find bounds for A_p and B_p . Following Hedberg (and borrowing an estimate for the Hardy-Littlewood maximal function from [48]) it is easy to find a constant for (71):

Theorem 3.16 [21] *Let $f \in L^p(\mathbb{R}^2)$, $1 < p < 2$, and define $\tilde{p}$ by $1/\tilde{p} = 1/p - 1/2$. Then we have the estimate*

$$\|\partial^{-1} f\|_{L^{\tilde{p}}} \leq A_p \|f\|_{L^p}$$

with the constant A_p given by

$$A_p = 2^{2-p/2} \left(\frac{25p}{p-1} \right)^{1/\tilde{p}} \max \left\{ 4\pi, \left(\frac{2\pi}{p'-2} \right)^{1/p'} \right\}. \quad (76)$$

For (74) we refer to a recent publication giving the best possible constants:

Theorem 3.17 [32] *Let $1 < p < \infty$. The norms of the two-dimensional Riesz transforms defined by (73) are*

$$\|R_j f\|_{L^p \rightarrow L^p} = B_p = \begin{cases} \tan(\pi/2p) & \text{if } 1 < p \leq 2, \\ \cot(\pi/2p) & \text{if } 2 \leq p < \infty. \end{cases} \quad (77)$$

3.2.2 Behaviour near $k = 0$ We state our theorem with assumptions similar to Nachman's. Note that for C^2 or smoother conductivities any ε strictly between 0 and 1 will do.

Theorem 3.18 *Let $\Omega \subset \mathbb{R}^2$ be a bounded, simply connected C^∞ domain. Let $\gamma \in W^{2,p}(\Omega)$, $1 < p < 2$, have a positive lower bound and assume $\gamma \equiv 1$ near $\partial\Omega$. With $\mathbf{t}(k)$ given in (57) and k close to zero we have*

$$|\mathbf{t}(k)| \leq C|k|^{1+\varepsilon} \quad \text{for all } 0 < \varepsilon < 2/p'.$$

Proof. We have $(\Lambda_\gamma - \Lambda_1)1 = 0$ and for any $f \in H^{1/2}(\partial\Omega)$ compute $\int_{\partial\Omega} \Lambda_\gamma f = \int_{\partial\Omega} \gamma \frac{\partial u}{\partial \nu} = \int_{\Omega} \nabla \cdot \gamma \nabla u = 0$; the same obviously holds for Λ_1 . Thus by (10) we have

$$\begin{aligned} |\mathbf{t}(k)| &= \left| \int_{\partial\Omega} (e^{i\bar{k}x} - 1)(\Lambda_\gamma - \Lambda_1)(\psi(\cdot, k) - 1) \right| \\ &\leq C \|e^{i\bar{k}x} - 1\|_{H^{1/2}(\partial\Omega)} \|\psi(\cdot, k) - 1\|_{H^{1/2}(\partial\Omega)}, \end{aligned}$$

where C is the operator norm of $\Lambda_\gamma - \Lambda_1$ from $H^{1/2}(\partial\Omega)$ to $H^{-1/2}(\partial\Omega)$.

For smooth functions $f, g : \partial\Omega \rightarrow \mathbb{C}$ we have by interpolation

$$\begin{aligned} \|fg\|_{H^{1/2}(\partial\Omega)} &\leq C \|fg\|_{L^2(\partial\Omega)} + C \|fg\|_{H^1(\partial\Omega)} \\ &\leq C \|f\|_{L^\infty(\partial\Omega)} \|g\|_{L^2(\partial\Omega)} + C (\|f\|_{L^\infty(\partial\Omega)} + \|f'\|_{L^\infty(\partial\Omega)}) \|g\|_{H^1(\partial\Omega)} \\ &\leq C \|f\|_{C^1(\partial\Omega)} \|g\|_{H^{1/2}(\partial\Omega)}. \end{aligned}$$

Since test functions are dense in $H^{1/2}(\partial\Omega)$ we get

$$\|\psi(\cdot, k) - 1\|_{H^{1/2}(\partial\Omega)} \leq C \|e^{ikx}\|_{C^1(\partial\Omega)} \|\mu(\cdot, k) - 1\|_{H^{1/2}(\partial\Omega)} + \|e^{ikx} - 1\|_{H^{1/2}(\partial\Omega)}.$$

Moreover, since $\|e^{ikx} - 1\|_{L^2(\partial\Omega)} \leq C \|e^{ikx} - 1\|_{L^\infty(\partial\Omega)}$ and $|(e^{ikx})'| \leq C|k|e^{|kx|}$ we have

$$\|e^{ikx} - 1\|_{H^{1/2}(\partial\Omega)} \leq C|k| \quad \text{for small } |k|.$$

Trace and Sobolev imbedding theorems combined with [43, Thm 3.3] and the fact that q is in our case compactly supported yield

$$\|\mu(\cdot, k) - 1\|_{H^{1/2}(\partial\Omega)} \leq C \|\mu(\cdot, k) - \sqrt{\gamma}\|_{H^1(\Omega)} \leq C \|\mu(\cdot, k) - \sqrt{\gamma}\|_{W_{-\beta}^{1,\bar{p}}} \leq C|k|^\varepsilon$$

for all $0 < \varepsilon < 2/p'$. Now we can estimate near $k = 0$

$$|\mathbf{t}(k)| \leq C|k|(C|k| + C|k|^\varepsilon) \leq C|k|^{1+\varepsilon}.$$

Q.E.D.

3.2.3 Symmetries

Theorem 3.19 *Let $\Omega \subset \mathbb{R}^2$ be a bounded, simply connected C^∞ domain. Let $\gamma_i \in C^2(\Omega)$ have a positive lower bound and satisfy $\gamma_i - 1 \in C_0^2(\Omega)$ for $i = 1, 2$. Set $q_i = \gamma_i^{-1/2} \Delta \gamma_i^{1/2}$ and denote by $\mathbf{t}_i$ the non-physical scattering transforms corresponding to q_i .*

(i) *Let $\varphi \in \mathbb{R}$ and denote $z_\varphi := e^{i\varphi}(z_1 + iz_2)$ for any $z \in \mathbb{R}^2$. Then the following identities are equivalent:*

$$\begin{aligned}\gamma_2(x) &= \gamma_1(x_{-\varphi}) & \text{for all } x \in \mathbb{R}^2; \\ \mathbf{t}_2(k) &= \mathbf{t}_1(k_\varphi) & \text{for all } k \in \mathbb{C} \setminus 0.\end{aligned}$$

(ii) *Let $\lambda > 0$. Then the following identities are equivalent:*

$$\begin{aligned}\gamma_2(x) &= \gamma_1(\lambda x) & \text{for all } x \in \mathbb{R}^2; \\ \mathbf{t}_2(k) &= \mathbf{t}_1(\lambda^{-1}k) & \text{for all } k \in \mathbb{C} \setminus 0.\end{aligned}$$

(iii) *The following identities are equivalent:*

$$\begin{aligned}\gamma_2(x) &= \gamma_1(-\bar{x}) & \text{for all } x \in \mathbb{R}^2; \\ \mathbf{t}_2(k) &= \overline{\mathbf{t}_1(\bar{k})} & \text{for all } k \in \mathbb{C} \setminus 0.\end{aligned}$$

(iv) *Fix an arbitrary $x' \in \mathbb{R}^2$. Then the following identities are equivalent:*

$$\begin{aligned}\gamma_2(x) &= \gamma_1(x - x') & \text{for all } x \in \mathbb{R}^2; \\ \mathbf{t}_2(k) &= e_k(x') \mathbf{t}_1(k) & \text{for all } k \in \mathbb{C} \setminus 0.\end{aligned}$$

Before proving the theorem we make some remarks.

Claim (i) of the theorem is analogous to the classical Karp's theorem: if the conductivity has rotational symmetry, so does the scattering transform.

Claim (ii) has an intuitive content: if the conductivity is pinched toward the origin so that details move deeper into Ω , the scattering transform decreases more slowly when $|k|$ grows; consequently the $\bar{\partial}$ equation is computationally harder to solve. This makes sense since in impedance tomography it is difficult to detect details from the center of the unknown body since the more spatial oscillation the fields have at the boundary, the less do currents enter into the middle.

The relation of Claim (iii) holds for the Fourier transform of real-valued functions. Since our potential q is always real, we see that the scattering transform requires a reflectional symmetry in addition.

Proof. (Theorem 3.19) Let us note that the existence of the scattering transforms $\mathbf{t}_i(k)$ follows from Theorem 3 in [43].

Sufficiency. Assume that $\gamma_1(x_{-\varphi}) = \gamma_2(x)$ and note that this implies $q_1(x_{-\varphi}) = q_2(x)$. We know from (24) that $g_{k_\varphi}(x) = g_k(x_\varphi)$ and we can compute

$$(g_{k_\varphi} * q_1)(x) = \int_{\mathbb{R}^2} g_k(x_\varphi - y_\varphi) q_1(y) dy = \rho_\varphi(g_k * q_2)(x),$$

where ρ_φ is the operator defined by $\rho_\varphi \psi(x) := \psi(x_\varphi)$. Moreover,

$$(g_{k_\varphi} * (q_1 \psi))(x) = \int_{\mathbb{R}^2} g_k(x_\varphi - z) q_2(z) \psi(z_{-\varphi}) dz = \rho_\varphi(g_k * (q_2 \cdot)) \rho_{-\varphi} \psi(x).$$

Noting that $\rho_\varphi^{-1} = \rho_{-\varphi}$ we get from (61)

$$\begin{aligned} \mu_1(x, k_\varphi) &= 1 - [I + g_{k_\varphi} * (q_1 \cdot)]^{-1} (g_{k_\varphi} * q_1) \\ &= 1 - \rho_\varphi [I + g_k * (q_2 \cdot)]^{-1} \rho_{-\varphi} \rho_\varphi (g_k * q_2) = \mu_2(x_\varphi, k). \end{aligned} \quad (78)$$

Substituting now (78) to the definition of $\mathbf{t}_1(k)$ yields

$$\mathbf{t}_1(k_\varphi) = \int_{\mathbb{R}^2} e_k(x_\varphi) q_2(x_\varphi) \mu_2(x_\varphi, k) dx = \mathbf{t}_2(k),$$

and sufficiency for (i) is proved.

For case (ii) note that the identity $\gamma_1(\lambda x) = \gamma_2(x)$ implies that

$$q_2(x) = \frac{\Delta \gamma_2^{1/2}}{\gamma_2^{1/2}}(x) = \frac{\Delta(\gamma_1^{1/2}(\lambda x))}{\gamma_1^{1/2}(\lambda x)} = \lambda^2 \frac{\Delta \gamma_1^{1/2}}{\gamma_1^{1/2}}(\lambda x) = \lambda^2 q_1(\lambda x)$$

for all $x \in \mathbb{R}^2$. We have $g_{\lambda k} = g_k(\lambda x)$ and thus $(g_k * q_1)(x) = m_{\lambda^{-1}}(g_{\lambda k} * q_2)(x)$, where $m_\alpha \psi(x) = \psi(\alpha x)$ for any $\alpha \in \mathbb{R}$. Similarly $g_k * (q_1 \cdot) = m_{\lambda^{-1}}(g_{\lambda k} * (q_2 \cdot)) m_\lambda$. We can deduce as earlier

$$\mu_1(x, k) = \mu_2(\lambda^{-1}x, \lambda k) \quad (79)$$

for all $x \in \mathbb{R}^2$ and $k \in \mathbb{C} \setminus 0$. This in turn yields $\mathbf{t}_2(\lambda k) = \mathbf{t}_1(k)$, and the sufficiency for case (ii) is proved.

By (25) we have $g_{\bar{k}}(x) = \overline{g_k(-\bar{x})}$ and thus, by the real-valuedness of $q_2(x) = q_1(-\bar{x})$ we can compute $(g_{\bar{k}} * q_1)(x) = R(g_k * q_2)(x)$ and similarly $g_{\bar{k}} * (q_1 \cdot) = R(g_k * (q_2 \cdot))R$, where $R\psi(x) = \psi(-\bar{x})$ with $R^{-1} = R$. We see as before that

$$\mu_1(x, \bar{k}) = \overline{\mu_2(-\bar{x}, k)} \quad (80)$$

for all $x \in \mathbb{R}^2$ and $k \in \mathbb{C} \setminus 0$. So $\overline{\mathbf{t}_2(k)} = \mathbf{t}_1(\bar{k})$, and the sufficiency for case (iii) is proved.

If $\gamma_1(x - x') = \gamma_2(x)$ we have $q_1(x - x') = q_2(x)$ and $(g_k * q_1)(x) = \tau_{x'}(g_k * q_2)(x)$ and similarly $g_k * (q_1 \cdot) = \tau_{x'}(g_k * (q_2 \cdot))\tau_{-x'}$, where $\tau_{x'}\psi(x) = \psi(x + x')$. We see that

$$\mu_1(x, k) = \mu_2(x + x', k) \quad (81)$$

for all $x \in \mathbb{R}^2$ and $k \in \mathbb{C} \setminus 0$, and so for any nonzero $k \in \mathbb{C}$ $\mathbf{t}_1(k) = e_k(-x')\mathbf{t}_2(k)$, and we are done with sufficiency.

Necessity. Assume that $\mathbf{t}_1(k_\varphi) = \mathbf{t}_2(k)$. Let $\gamma_3(x) = \gamma_1(x_{-\varphi})$ and denote by $\mathbf{t}_3(k)$ the non-physical scattering transform of γ_3 . By the sufficiency proof we know that $\mathbf{t}_3(k) = \mathbf{t}_1(k_\varphi) = \mathbf{t}_2(k)$ and by the uniqueness of Nachman's reconstruction method $\gamma_2(x) = \gamma_3(x) = \gamma_1(x_{-\varphi})$.

The cases (ii),(iii) and (iv) can be proved similarly to case (i).

Q.E.D.

4. Computational methods

4.1. Evaluating Faddeev Green's functions

To evaluate the function $g_k(x)$ for $x \in \mathbb{R}^2 \setminus 0$ and $k \in \mathbb{C} \setminus 0$ it is enough by Theorem 3.2 to know $g_1(x)$ for $x_1 > 0$, since $g_k(x) = g_1(kx)$ and $g_1(-\bar{x}) = \overline{g_1(x)}$.

4.1.1 Computing g_1 for $|x| > 1/2$ We will present an algorithm for $g_1(x)$ containing only one-dimensional exponentially converging integrals in addition to explicit formulas.

Start by choosing some $N > 0$ for the formula

$$g_1(x) = \frac{e^{-ix_1}}{2\pi} \operatorname{Re} \left[-e^{ix_1} \sum_{j=0}^N \frac{j!}{(ix)^{j+1}} + \frac{(N+1)!e^{ix_1}}{(-x)^{N+1}} \int_0^\infty \frac{e^{-t(x_1+ix_2)}}{(t-i)^{N+2}} dt \right] \quad (82)$$

given by Theorem 3.11. This is suitable for numerical work for $x_1 > 0$ not too close to zero since the integrand is bounded by $e^{-x_1 t}$.

However, the integral appearing in Formula (82) can be easily modified to give better convergence properties for small x_1 , too. We have in fact seen this already in the proof of Theorem 3.8. Namely, with similar arguments we get

$$\int_0^\infty \frac{e^{-ix_2 t - x_1 t}}{(t-i)^{N+2}} dt = -i \int_0^\infty \frac{e^{-x_2 s + ix_1 s}}{(-is-i)^{N+2}} ds \quad (83)$$

for $x_2 > 0$ and $x_1 \geq 0$; note that the case $x_1 = 0$ is included. For $x_2 < 0$ and $x_1 \geq 0$ compute in a similar fashion

$$\int_0^\infty \frac{e^{-ix_2 t - x_1 t}}{(t-i)^{N+2}} dt = (1+i) \int_0^\infty \frac{e^{-is(x_2+x_1)+s(x_2-x_1)}}{(s+is-i)^{N+2}} ds. \quad (84)$$

Formulas for $x_1 < 0$ follow by reflection.

4.1.2 Computing g_1 in the unit disc Recall from Section 3.1.1 that g_1 can be written in the form

$$g_1(x) = e^{-ix}H_1(x) - \frac{e^{-ix}}{2\pi} \log|x|, \quad (85)$$

where H_1 is a harmonic function by Lemma 3.1. With the values $g_1(x)$ for $|x| = 1$ available from the algorithm of Section 4.1.1, we can first find H_1 on the unit circle from

$$H_1(x) = e^{ix}g_1(x), \quad |x| = 1.$$

Then we use the classical Poisson kernel to compute $H_1(x)$ inside the unit disc:

$$H_1(x) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \frac{1-r^2}{1-2r\cos(\theta-t)+r^2} H_1(e^{it}) dt, \quad (86)$$

where $x = re^{i\theta}$, $r < 1$. Now compute g_1 from (85) for $0 < |x| < 1$.

This procedure has an additional advantage: when implementing numerically convolution integrals containing the function g_1 , equation (85) makes it possible to deal explicitly with the singular part. Note also that by Lemma 3.12 we know that $H_1(0) = -\frac{\gamma_0}{2\pi}$, so it is possible to check the accuracy of the numerical algorithm for H_1 near the origin.

The error of this strategy is controlled by the maximum principle: suppose we have a numerically computed approximation $\tilde{g}_1(x)$ for $g_1(x)$:

$$g_1(x) = \tilde{g}_1(x) + E(x), \quad x \in \mathbb{R}^2, \quad |x| = 1.$$

Denote by $\tilde{H}_1(x)$ the function defined by (86) with $\tilde{g}_1(x)$ in place of $g_1(x)$. Then

$$\|H_1 - \tilde{H}_1\|_{L^\infty(|x| \leq R)} \leq e \sup_{|x|=1} |E(x)|.$$

4.1.3 Numerical results The code computing Faddeev Green's function $g_1(x)$ is simple: choose first a radius $1/2 < R < 1$. This is necessary since the Poisson kernel resembles the Dirac delta near the boundary $|x| = 1$ and becomes numerically inaccurate to compute. Divide the plane to seven domains according to Figure 4. In domains 3 and 6 we use formula (82), in domains 4 and 7 formula (83) and in domains 2 and 5 formula (84). In domain 1 we use the Poisson approach of Section 4.1.2.

In practice we chose $R = 4/5$. Only 200 values on equidistant points of S^1 were needed to give relative error 10^{-9} in the numerical integration of the Poisson kernel in domain 1. In the other 6 domains, choosing a suitable amount of Gauss-Legendre quadrature points for the exponentially convergent integral gave also relative error 10^{-9} . This was checked by changing the order N of the asymptotic expansion (82) and making sure that the change is within the accepted error. Due to the simplicity of the algorithm the code is fast.

Plots of the real and imaginary parts of the Faddeev Green's function $g_1(x)$ are given in Figure 5. Figure 6 shows the real harmonic function H_1 .

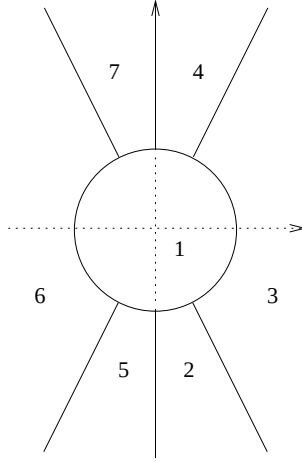


Figure 4: The seven planar domains having distinct computational strategies. The slopes of the skew lines appearing in the picture are either 2 or -2 . The circle has radius R .

4.1.4 An alternative algorithm for g_1 Before finding the above method based on residue calculus we implemented another algorithm. Although this approach is clearly worse (relative error at best 10^{-5} and computation times unacceptably long due to the expensive 2D integrations) we describe it briefly.

The integrand in

$$\int \frac{e^{ix \cdot \xi}}{\xi(\bar{\xi} + 2)} d\xi \quad (87)$$

has singularities at $\xi = 0$ and $\xi = -2$; we will treat those points separately. For this, choose a constant $0 < \alpha < 1/2$ and define

$$R_\alpha^+ = 2 + 2\alpha, \quad R_\alpha^- = 2 - 2\alpha. \quad (88)$$

Write the integral expression for $g_1(x)$ in four parts analogously to Lemma 3.4 of [43]:

$$\begin{aligned} (2\pi)^2 g_1(x) = & \int_{|\xi| < R_\alpha^+} \frac{-2(e^{ix \cdot \xi} - 1)}{|\xi|^2(\bar{\xi} + 2)} d\xi + \int_{|\xi| > R_\alpha^+} \frac{-2e^{ix \cdot \xi}}{|\xi|^2(\bar{\xi} + 2)} d\xi \\ & + \int_{|\xi| < R_\alpha^+} \frac{d\xi}{\xi(\bar{\xi} + 2)} + \left\{ \int_{|\xi| < R_\alpha^+} \frac{e^{ix \cdot \xi} - 1}{|\xi|^2} d\xi + \int_{|\xi| > R_\alpha^+} \frac{e^{ix \cdot \xi}}{|\xi|^2} d\xi \right\}. \end{aligned} \quad (89)$$

Combining (89) with slightly modified [43, Lemma 3.5] we get

$$g_1(x) = -\frac{\gamma_0}{2\pi} - \frac{1}{2\pi} \log \frac{R_\alpha^+ |x|}{2} + \frac{1}{2\pi} \log(1 + \alpha)$$

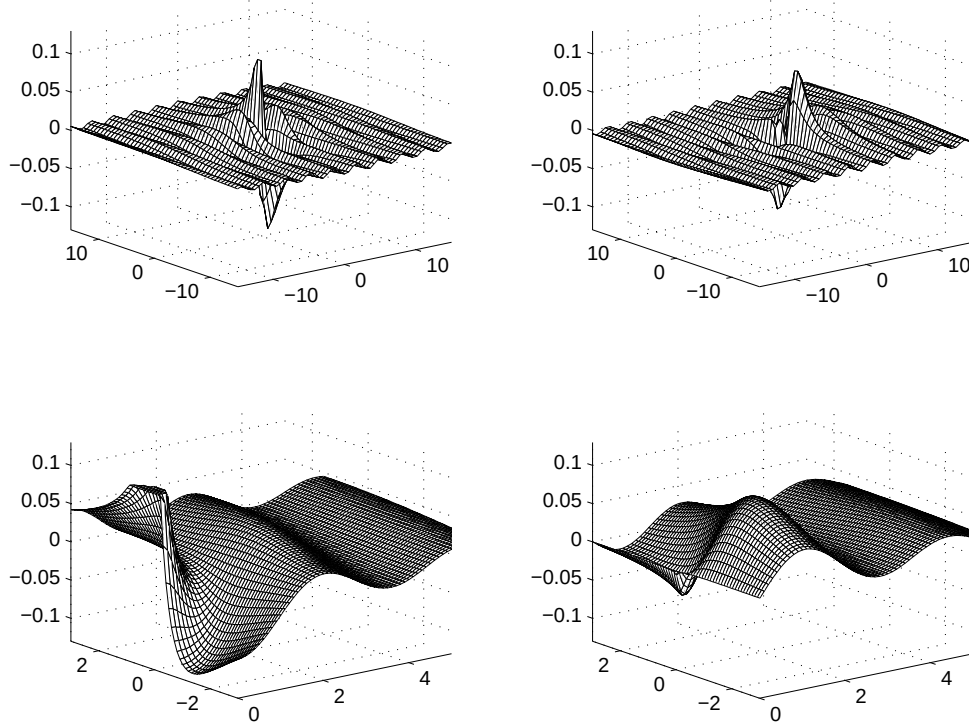


Figure 5: In the left column are plots of the real part of Faddeev's fundamental solution $g_1(x)$. The positive spike is the logarithmic singularity and the negative spike just a local minimum. In the lower plot we take a closer look at the real part near the origin. For clarity, the picture is cut along the imaginary axis; the function will continue in an even fashion to the left half-plane. In the right column are plots of the imaginary part of $g_1(x)$. Again, the lower plot is a closer look near the origin. Note that the function will continue in an odd fashion to the left half-plane.

$$-\frac{1}{2\pi^2} \underbrace{\int_{|\xi| < R_\alpha^+} \frac{e^{ix \cdot \xi} - 1}{|\xi|^2(\bar{\xi} + 2)} d\xi}_{I_1} - \frac{1}{2\pi^2} \underbrace{\int_{|\xi| > R_\alpha^+} \frac{e^{ix \cdot \xi}}{|\xi|^2(\bar{\xi} + 2)} d\xi}_{I_2}.$$

We have to design numerical algorithms for the computation of I_1 and I_2 .

The integrand of I_1 has two singular points, namely the origin and the point $\xi = -2$. To be able to deal with these points separately we divide the disc $|\xi| < R_\alpha^+$ to three domains $D_j, j = 1, 2, 3$, given by

$$\begin{aligned} D_1 &= \{(r, \phi) \mid 0 < r < R_\alpha^-, 0 < \phi < 2\pi\}, \\ D_2 &= \{(r, \phi) \mid R_\alpha^- < r < R_\alpha^+, -\pi + \kappa_0 < \phi < \pi - \kappa_0\}, \\ D_3 &= \{(r, \phi) \mid R_\alpha^- < r < R_\alpha^+, \pi - \kappa_0 < \phi < \pi + \kappa_0\}, \end{aligned} \tag{90}$$

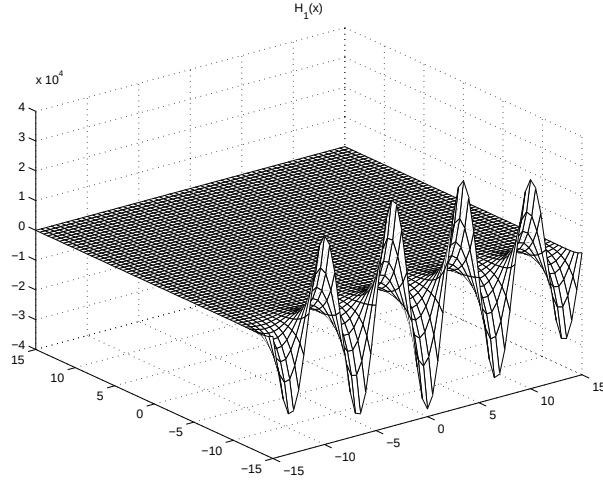


Figure 6: The real-valued harmonic function $H_1(x) = G_1(x) - G_0(x)$.

where $\kappa_0 = 1/2 \arccos(1 - \alpha)$ is a constant angle. This choice of κ_0 ensures that D_3 is starlike with respect to the point $(-2, 0)$; see Figure 7.

In D_1 and D_2 , choosing polar coordinates r, ϕ centered at the origin yields a numerically feasible integration strategy. In D_3 , however, there is a singularity at the point $\xi = -2$. Translating this point to the origin with the coordinate change $\zeta = \xi + 2$ and writing the integral in polar coordinates removes this difficulty. A somewhat involved piecewise formula for the boundary $r(\theta)$ can be found with elementary geometry.

The evaluation of I_2 can be done using asymptotic expansions of Bessel functions. Recall that $|\bar{\xi}| > 2$ and write I_2 for any $N > 0$ in the form

$$\int_{|\xi| > R_\alpha^+} \frac{e^{ix \cdot \xi}}{|\xi|^2 \bar{\xi}} \frac{d\xi}{(1 - (-2/\bar{\xi}))} = \sum_{\nu=1}^{N+1} \int_{|\xi| > R_\alpha^+} \frac{(-2)^{\nu-1} e^{ix \cdot \xi}}{|\xi|^2 \bar{\xi}^\nu} d\xi + \int_{|\xi| > R_\alpha^+} \frac{(-2)^{N+1} e^{ix \cdot \xi}}{|\xi|^2 \bar{\xi}^{N+1} (\bar{\xi} + 2)} d\xi,$$

where the first term is easy to integrate numerically with large enough N . The integrals in the second term can be written in a convenient form using Bessel functions: denote $\arg(x) = \theta_0$ and compute

$$\begin{aligned} \int_{|\xi| > R_\alpha^+} \frac{e^{ix \cdot \xi}}{|\xi|^2 \bar{\xi}^\nu} d\xi &= 2\pi e^{i\nu(\theta_0 + \pi/2)} \int_{R_\alpha^+}^\infty \frac{1}{r^{\nu+1}} \left(\frac{1}{2\pi} \int_0^{2\pi} e^{i|x|r \cos \varphi} e^{i\nu(\varphi - \pi/2)} d\varphi \right) dr \\ &= 2\pi (ix)^\nu \int_{R_\alpha^+|x|}^\infty \frac{J_\nu(t)}{t^{\nu+1}} dt, \end{aligned} \quad (91)$$

where we used a formula from Section 3.6.1 in [41]. Now a numerically reasonable scheme to compute the integrals (91) is achieved by using the asymptotic expansion given as Formulae 1, 7 and 8 in Section 8.451 of [19].

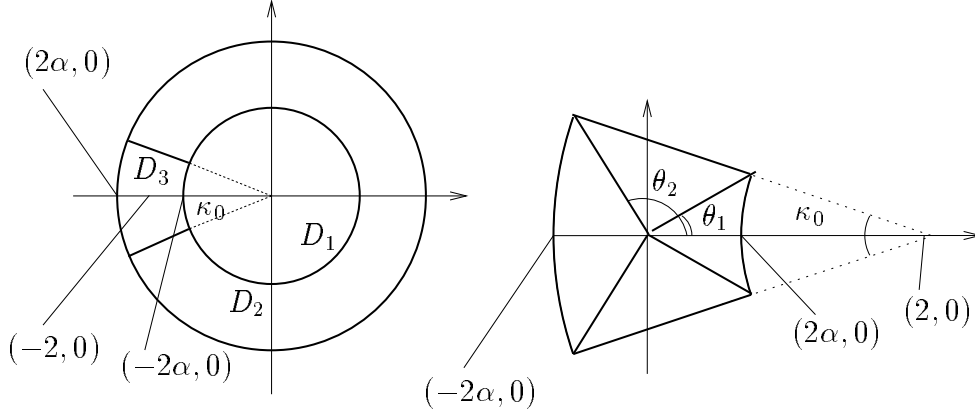


Figure 7: The division of the closed disc of radius R_α^+ to three domains of integration is shown on the left. On the right we see the polar coordinates chosen for D_3 .

For convenience, we summarize the two-dimensional method as

$$\begin{aligned}
g_1(x) = & -\frac{\gamma_0}{2\pi} - \frac{1}{2\pi} \log \frac{R_\alpha^+ |x|}{2} + \frac{1}{2\pi} \log(1 + \alpha) \\
& - \frac{1}{2\pi^2} \left[\int_0^{2\pi} \int_0^{R_\alpha^-} \frac{e^{i(x_1 r \cos \phi + x_2 r \sin \phi)} - 1}{r(re^{-i\phi} + 2)} dr d\phi \right. \\
& + \int_{-\pi + \kappa_0}^{\pi - \kappa_0} \int_{R_\alpha^-}^{R_\alpha^+} \frac{e^{i(x_1 r \cos \phi + x_2 r \sin \phi)} - 1}{r(re^{-i\phi} + 2)} dr d\phi \\
& + \int_0^{2\pi} \int_0^{\tilde{r}(\theta)} \frac{e^{i(x_1 \tilde{r} \cos \theta + x_2 \tilde{r} \sin \theta - 2x_1)} - 1}{|\tilde{r}e^{i\theta} - 2|^2 e^{-i\theta}} d\tilde{r} d\theta \\
& + (-2)^{N+1} \int_0^{2\pi} \int_{R_\alpha^+}^\infty \frac{e^{i(x_1 r \cos \phi + x_2 r \sin \phi + (N+1)\phi)}}{r^{N+2}(re^{-i\phi} + 2)} dr d\phi \\
& \left. - \pi \sum_{\nu=1}^{N+1} (-2ix)^\nu \int_{R_\alpha^+ |x|}^\infty \frac{J_\nu(t)}{t^{\nu+1}} dt \right],
\end{aligned}$$

where $N > 1$ is an integer.

4.2. Computing μ and $\mathbf{t}$ from γ

When solving the inverse conductivity problem practically with Nachman's method, it is important to be able to compute the functions $\mu(x, k)$ and $\mathbf{t}(k)$ for some known conductivities. This enables us to test the intermediate results in the reconstruction.

4.2.1 Finding μ in a bounded domain For a given conductivity γ , we are interested in finding the functions $\mu(x, k)$ for x ranging in a neighbourhood of $\text{supp}(q)$ and k belonging to a finite collection $K \subset \mathbb{C}$ of nonzero complex numbers. With this information we are able to

- compute $\mu(x, k)$ for $k \in K$ and any $x \in \mathbb{R}^2$ since

$$\mu(x, k) = 1 - g_k * (q(\cdot)\mu(\cdot, k)),$$

where on the right hand side only the values of $\mu(\cdot, k)$ on $\text{supp}(q)$ are needed; in particular we can examine the boundary values of the Faddeev solutions $\psi(x, k)|_{\partial\Omega}$ appearing in formulas (9) and (10) because

$$\psi(x, k) = e^{ikx}\mu(x, k);$$

- compute the non-physical scattering transform $\mathbf{t}(k)$ for all $k \in K$ from

$$\mathbf{t}(k) := \int_{\mathbb{R}^2} e_k(x)\mu(x, k)q(x)dx; \quad (92)$$

- test any numerical solution of the $\bar{\partial}$ equation (11);
- observe the rate of convergence $\mu(x, k) \rightarrow \sqrt{\gamma(x)}$ when $k \rightarrow 0$.

For this purpose we will prove that to find the restriction of $\mu(\cdot, k)$ on Ω it is enough to solve an analogue of the Lippmann-Schwinger equation on a slightly larger set than $\text{supp}(q)$.

Theorem 4.1 *Let $\Omega \subset \mathbb{R}^2$ be a bounded, simply connected domain with smooth boundary. Let $\gamma \in C^\infty(\mathbb{R}^2)$ have a positive lower bound and satisfy $\gamma - 1 \in C_0^\infty(\Omega)$, and set $q = \gamma^{-1/2}\Delta\gamma^{1/2}$. For any $k \in \mathbb{C} \setminus 0$ denote by $\mu(x, k)$ the unique smooth solution of*

$$(-\Delta - 4ik\bar{\partial} + q)\mu(\cdot, k) = 0 \quad \text{in } \mathbb{R}^2$$

satisfying $\mu(\cdot, k) - 1 \in W^{1, \tilde{p}}$ for some $1 < p < 2, 1/\tilde{p} = 1/p - 1/2$. The solution exists by Theorem 3.15.

Take any bounded domain $D \subset \mathbb{R}^2$ with $\text{supp}(q) \subset D$. Then the operator

$$I + g_k * (q \cdot) : W^{1, \tilde{p}}(D) \rightarrow W^{1, \tilde{p}}(D) \quad (93)$$

is invertible. Moreover,

$$\mu(x, k)|_D = [I + g_k * (q \cdot)]^{-1}1. \quad (94)$$

Proof. Note the decomposition

$$W^{1,\tilde{p}}(D) \xrightarrow{\iota} C(\overline{D}) \xrightarrow{q\cdot} L^p \xrightarrow{g_k^*} W^{1,\tilde{p}} \rightarrow W^{1,\tilde{p}}(D),$$

where ι is (compact) Sobolev imbedding. Thus (93) is Fredholm of index zero.

We have to show that (93) is one-to-one. Let $u \in W^{1,\tilde{p}}(D)$ satisfy

$$u + g_k * (qu) = 0 \quad \text{in } D. \quad (95)$$

Extend u to the whole plane by $u = -g_k * (qu)$; by [43, Lemma 1.3] we have $u \in W^{1,\tilde{p}}$. Clearly u satisfies $(-\Delta - 4ik\bar{\partial})u = -qu$, and by [43, Lemma 1.5] we see that $u \equiv 0$ in $\mathbb{R}^2$, as wished. Finally note that the difference of the functions in (94) satisfy (95), so they must coincide. **Q.E.D.**

4.2.2 Description of the algorithm To implement Theorem 4.1 practically we have to construct a computational framework. We will find a finite approximate basis for $W^{1,\tilde{p}}(D)$ and approximate the operator $g_k * (q\cdot)$ with a matrix acting on the subspace spanned by our basis functions. This projection method is described generally in Chapter 4 of [36] and Chapter 13 of [37]. We present here in detail the method and a proof of its convergence.

Given q as in the assumptions of Theorem 4.1, let $d > 0$ be so large that $\text{supp}(q) \subset D$ if D is the square given by

$$D := \{x \in \mathbb{R}^2 \mid -d < x_i < d, i = 1, 2\}. \quad (96)$$

For any $N \geq 2$ denote $h = 2d/(N-1)$ and define a finite grid $G_N \subset \overline{D}$ by

$$G_N = \{(x_1, x_2) \in \overline{D} \mid x_i = -d + \nu h, \nu = 0, \dots, N-1, i = 1, 2\}. \quad (97)$$

Let $x^{(j)}, j = 1, \dots, N^2$, be a numbering of the points in G_N and define a linear subspace $V_N \subset W^{1,\tilde{p}}(D)$ by

$$V_N = \text{span}\{v_j \mid j = 1, \dots, N^2\}, \quad (98)$$

where the continuous, piecewise linear basis functions $v_j : D \rightarrow \mathbb{R}$ are defined by the requirements

- (i) $v_j(x^{(k)}) = \delta_{jk}$ for $j, k = 1, \dots, N^2$;
- (ii) v_j is linear on all triangular subsets of D having vertices at grid points, two sides parallel to the coordinate axis and third side having slope 1.

It is clear that $V_N \subset W^{1,\tilde{p}}$ since the distribution derivatives of functions in V_N are piecewise constant and thus integrable to any power.

Since $W^{1,\tilde{p}}(D)$ functions are continuous and bounded on $\overline{D}$ by the Sobolev imbedding theorem, we are able to define a continuous projection

$$P_N : W^{1,\tilde{p}}(D) \rightarrow V_N \quad (99)$$

by the formula $P_N(f) = \sum_{j=1}^{N^2} f(x_j)v_j(x)$. We will approximate the operator $g_k * (q \cdot)$ with a matrix $P_N(g_k * (q \cdot))$ acting on V_N , a finite-dimensional subspace¹. It turns out that $P_N(I + g_k * (q \cdot))$ is invertible if N is taken large enough and that

$$\lim_{N \rightarrow \infty} [P_N(I + g_k * (q \cdot))]^{-1} 1 = [I + g_k * (q \cdot)]^{-1} 1 = \mu(\cdot, k)|_D$$

in the $W^{1,\tilde{p}}(D)$ topology and thus we have pointwise convergence on $\overline{D}$. We postpone the proof of these facts to Section 4.2.3.

Now a numerical algorithm for the computation of $\mu(\cdot, k)$ for a fixed $k \in \mathbb{C} \setminus 0$ can be realized as follows:

1. Pick an initial $N \geq 2$.
2. Compute the $N^2 \times N^2$ matrix $A_N = (a_N^{ij})$, where

$$a_N^{ij} := (g_k * (qv_j))(x^{(i)}) = \int_D g_k(x^{(i)} - y)q(y)v_j(y)dy. \quad (100)$$

This requires just 2D numerical integration over the support of a basis function v_j ; note that in the case $\text{supp}(q) \cap \text{supp}(v_j) = \emptyset$ the matrix element is readily seen to vanish. Writing the integration in polar coordinates eliminates the numerical problem of the logarithmic singularity of $g_k(x - y)$ at $x = y$.

3. If the matrix $P_N(I + A) = I + A_N$ (acting in the finite-dimensional space V_N) is not invertible, go back to Step 2 with a larger N . Eventually N is so large that $(I + A_N)^{-1}$ exists and we can compute it numerically.
4. Find approximate point values $\mu_N(x^{(j)}, k)$ of $\mu(\cdot, k)$ at the grid points:

$$\begin{bmatrix} \mu_N(x^{(1)}, k) \\ \vdots \\ \mu_N(x^{(N^2)}, k) \end{bmatrix} = (I + A_N)^{-1} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}. \quad (101)$$

¹Since $W^{1,\tilde{p}}(D)$ is not a Hilbert space, this is a sensitive step. Namely, not in all Banach spaces is it possible to approximate compact operators with finite rank ones as shown by the examples of Enflo [13] and Szankowski [52]. In the latter paper the counterexample is $L(H)$, the space of bounded linear operators on a Hilbert space H . In our Banach space $W^{1,\tilde{p}}(D)$ we fortunately do not have to deal with this problem.

4.2.3 Proof of convergence Throughout this section we keep $k \in \mathbb{C} \setminus 0$ fixed and denote

$$A := g_k * (q \cdot) : W^{1,\tilde{p}}(D) \rightarrow W^{1,\tilde{p}}(D), \quad (102)$$

$$A_N := P_N A : W^{1,\tilde{p}}(D) \rightarrow V_N. \quad (103)$$

Theorem 4.2 *With the assumptions of Theorem 4.1, let $k \in \mathbb{C} \setminus 0$ and define d , D , h , G_N , V_N and P_N for any $N \geq 2$ as in Section 4.2.2.*

Then the matrix

$$P_N(I + g_k * (q \cdot)) : V_N \rightarrow V_N \quad (104)$$

is invertible for N large enough and, in the $C(\overline{D})$ topology,

$$\lim_{N \rightarrow \infty} [P_N(I + g_k * (q \cdot))]^{-1} 1 = [I + g_k * (q \cdot)]^{-1} 1. \quad (105)$$

Before proving Theorem 4.2 we need some lemmas.

Lemma 4.3 *With the assumptions of Theorem 4.2 we have for a fixed $u \in C^1(\overline{D})$*

$$\|P_N u - u\|_{W^{1,\tilde{p}}(D)} \rightarrow 0 \text{ when } N \rightarrow \infty.$$

Moreover, there exists a uniform bound

$$\|P_N\|_{C^1(\overline{D}) \rightarrow W^{1,\tilde{p}}(D)} \leq C_P < \infty \quad \text{for all } N \geq 2. \quad (106)$$

Proof. Fix a function $u \in C^1(\overline{D})$; with no loss of generality we can take u to be real-valued. Denote $u_N = P_N u$ for any $N \geq 2$.

Let $\varepsilon > 0$. Our aim is to find such an $N_\varepsilon \in \mathbb{N}$ that

$$\sum_{|\alpha| \leq 1} \|\partial^\alpha (u_N - u)\|_{L^{\tilde{p}}(D)} < \varepsilon \quad \text{for all } N \geq N_\varepsilon. \quad (107)$$

Let us take first the case $\alpha = 0$. For an $N \geq 2$, pick any open triangle T belonging to the triangulation of $\overline{D}$ corresponding to N . We want to bound the function $|u - u_N|$ on T and for this we need an estimate on its gradient. First of all, since $u \in C^1(\overline{D})$ we have

$$\max_{x \in \overline{T}} |\nabla \tilde{u}(x)| \leq 2\|u\|_{C^1(\overline{D})}.$$

Furthermore, we can estimate on T the gradient ∇u_N (which is constant) by

$$|\nabla u_N(x)| \leq \frac{\sqrt{2}}{h} \max_{x', x'' \in \nu(T)} |u(x') - u(x'')| \leq \frac{\sqrt{2}}{h} \sqrt{2} h \sup_{x \in T} |\nabla u(x)| \leq 4\|u\|_{C^1(\overline{D})},$$

where $\nu(T)$ is the set of vertices of T .

The function $u - u_N$ is continuous on $\overline{T}$, vanishes at the vertices of T and satisfies $|\nabla(u - u_N)| \leq 6\|u\|_{C^1(\overline{D})}$ on T . Estimate

$$|u(x) - u_N(x)| \leq 6\|u\|_{C^1(\overline{D})}h < C/(N - 1) < \varepsilon/(3|D|^{1/\tilde{p}})$$

for $x \in T$ and N large enough. Thus

$$\|u - u_N\|_{L^{\tilde{p}}(D)} = \left(\sum_T \int_T |u - u_N|^{\tilde{p}}\right)^{1/\tilde{p}} < \varepsilon/3,$$

and the first term of (107) is under control.

Let us consider the partial derivative $(\partial/\partial x_1)u = \partial_1 u$. Since $\partial_1 u$ is uniformly continuous on the compact set $\overline{D}$, there exists such a $\delta > 0$ that

$$\partial_1 u(B(x, \delta)) \subset B(\partial_1 u(x), \varepsilon/(3|D|^{1/\tilde{p}}))$$

for all $x \in \overline{D}$. Enlarge N_ε (if necessary) to satisfy $N_\varepsilon > 4d/\delta + 1$. Let $N \geq N_\varepsilon$ and pick any open triangle T belonging to the triangulation of $\overline{D}$ corresponding to N . It follows that $T \subset B(x, 2h) \subset B(x, \delta)$ for any $x \in \overline{T}$.

Note that the piecewise linearity of u_N implies that $a := \partial_1 u_N$ is constant on T . Furthermore, on the horizontal side of T there necessarily is a point $x \in \overline{T}$ with $\partial_1 u(x) = a$. Thus

$$(\partial_1(u - u_N))(T) \subset (\partial_1(u - u_N))(B(x, \delta)) \subset B(0, \varepsilon/(3|D|^{1/\tilde{p}})),$$

and we get

$$\|\partial_1(u - u_N)\|_{L^{\tilde{p}}(D)} = \left(\sum_T \int_T |\partial_1(u - u_N)|^{\tilde{p}}\right)^{1/\tilde{p}} < \varepsilon/3,$$

so we are done with the second term of (107).

A similar argument takes care of the x_2 -derivative, and the first claim is proved.

To prove (106) note that the above implies for a fixed $u \in C^1(\overline{D})$

$$\|P_N u\|_{W^{1,\tilde{p}}(D)} \leq \|u\|_{W^{1,\tilde{p}}(D)} + \|P_N u - u\|_{W^{1,\tilde{p}}(D)} \leq \|u\|_{C^1(\overline{D})} < \infty$$

for all $N \geq 2$. Now an application of the Banach-Steinhaus theorem completes the proof. **Q.E.D.**

Lemma 4.4 *With the assumptions of Theorem 4.2 and A, A_N given by (102) and (103) we have*

$$\lim_{N \rightarrow \infty} A_N = A \tag{108}$$

in the operator topology $W^{1,\tilde{p}}(D) \rightarrow W^{1,\tilde{p}}(D)$.

Proof. Let $\varepsilon > 0$ and $\|f\|_{W^{1,\tilde{p}}(D)} \leq 1$; denote $u = Af$. The decomposition

$$W^{1,\tilde{p}}(D) \xrightarrow{q} W^{1,\tilde{p}}(D) \rightarrow W^{1,p}(D) \xrightarrow{g_k^*} W^{2,\tilde{p}}(D) \xrightarrow{\iota} C^1(\overline{D})$$

allows us to conclude that $\overline{A(B)} \subset C^1(\overline{D})$ is a compact set, where $B = B(0, 1)$ is the unit ball of $W^{1,\tilde{p}}(D)$.

Choose finitely many functions $u_j \in \overline{A(B)}$ that satisfy in $C^1(\overline{D})$

$$\overline{A(B)} \subset \bigcup_j B(u_j, \varepsilon/2(C_P + C_I)),$$

where C_P is the uniform bound for the operator norms P_N given in Lemma 4.3 and C_I is the finite operator norm of the inclusion map $C^1(\overline{D}) \rightarrow W^{1,\tilde{p}}(D)$. By Lemma 4.3 there exists, for the finite collection $\{u_j\}$, a natural number N_ε such that

$$\|P_N u_j - u_j\|_{W^{1,\tilde{p}}(D)} < \varepsilon/2 \quad (109)$$

for all j and $N \geq N_\varepsilon$. Now for some j it is possible to write $Af = u = u_j + (u - u_j)$ with

$$\|u - u_j\|_{C^1(D)} < \varepsilon/2(C_P + C_I), \quad (110)$$

and we get from (109) and (110)

$$\begin{aligned} \|(I - P_N)Af\|_{W^{1,\tilde{p}}(D)} &= \|(I - P_N)u\|_{W^{1,\tilde{p}}(D)} \leq \\ &\leq \|(I - P_N)u_j\|_{W^{1,\tilde{p}}(D)} + \|I - P_N\|_{C^1(\overline{D}) \rightarrow W^{1,\tilde{p}}(D)} \|(u - u_j)\|_{C^1(\overline{D})} < \varepsilon \end{aligned}$$

for $N \geq N_\varepsilon$.

Q.E.D.

Proof. (Theorem 4.2) Recall that by Theorem 4.1 the operator $I + A$ is invertible on $W^{1,\tilde{p}}(D)$. Thus there exists such a $c > 0$ that we can estimate for any $f \in V_N \subset W^{1,\tilde{p}}(D)$

$$\begin{aligned} \|P_N(I + A)f\|_{V_N} &= \|(I + A_N)f\|_{W^{1,\tilde{p}}(D)} \\ &\geq \|(I + A)f\|_{W^{1,\tilde{p}}(D)} - \|(A_N - A)f\|_{W^{1,\tilde{p}}(D)} \\ &\geq c\|f\|_{V_N} - \|A_N - A\|_{L(W^{1,\tilde{p}}(D))} \|f\|_{V_N}. \end{aligned} \quad (111)$$

By Lemma 4.4 we know that $\|A_N - A\|_{L(W^{1,\tilde{p}}(D))} < c$ for large enough N , and (111) implies that for those N the matrix $P_N(I + A) : V_N \rightarrow V_N$ is invertible.

To prove the second claim, note first that by Lemma 4.4 the operator

$$I + A_N = (I + A) + (A_N - A) : W^{1,\tilde{p}}(D) \rightarrow W^{1,\tilde{p}}(D)$$

is invertible for large enough N since the set of invertible operators is open in $L(W^{1,\tilde{p}}(D))$. Moreover, because taking inverse element is a continuous operation in any Banach algebra we have

$$\lim_{N \rightarrow \infty} (I + A_N)^{-1} = (I + A)^{-1} \quad (112)$$

in the $L(W^{1,\tilde{p}}(D))$ topology.

Finally we check that

$$[P_N(I + A)]^{-1}1 = (I + A_N)^{-1}1. \quad (113)$$

We claim that $(I + A_N)^{-1}1 \in V_N$. If this were not the case, there would be an $f \in W^{1,\tilde{p}}(D) \setminus V_N$ satisfying $(I + A_N)f = 1$, which implies $f = 1 - A_N f \in V_N$, a contradiction. Thus we have

$$0 = (P_N - I)(I + A_N)^{-1}1 = (P_N - I + I + A_N)(I + A_N)^{-1}1 - 1 = P_N(I + A)(I + A_N)^{-1}1 - 1,$$

which gives (113). Combining now (112) and (113) completes the proof. **Q.E.D.**

4.2.4 Generating test problems We will construct building blocks for smooth test conductivities and potentials. Let $0 < d < 1$ and define a non-negative function $\Psi_d \in C_0^\infty((-1, 1))$ by

$$\Psi_d(t) = \begin{cases} e^{-\frac{2(d^2+t^2)}{(t+d)^2(t-d)^2}} & \text{for } -d \leq t \leq d, \\ 0 & \text{otherwise.} \end{cases}$$

To use Ψ_d in the construction of test conductivities on the unit disc, compute in polar coordinates

$$\Delta \Psi_d(r) = \frac{\partial^2 \Psi_d(r)}{\partial r^2} + \frac{1}{r} \frac{\partial \Psi_d(r)}{\partial r},$$

where

$$\begin{aligned} \frac{\partial \Psi_d(r)}{\partial r} &= -\Psi_d(r) \frac{4r(3d^2 + r^2)}{(d+r)^3(d-r)^3}, \\ \frac{\partial^2 \Psi_d(r)}{\partial r^2} &= \Psi_d(r) \left[\frac{16r^2(3d^2 + r^2)^2}{(d+r)^6(d-r)^6} - \frac{24r^2(3d^2 + r^2)}{(d+r)^4(d-r)^4} - \frac{12(d^2 + r^2)}{(d+r)^3(d-r)^3} \right]. \end{aligned}$$

Define a radially symmetric conductivity $\gamma \in C^\infty(\mathbb{R}^2)$ by

$$\gamma(r) := (\alpha \Psi_d(r) + 1)^2, \quad (114)$$

where $\alpha > -\exp(2/d^2)$ is a real constant. Note that $\text{supp}(\gamma - 1) = \overline{B(0, d)}$ and that γ is bounded away from zero. The Schrödinger potential q corresponding to γ is

$$q(r) = \frac{\Delta \gamma^{1/2}(r)}{\gamma^{1/2}(r)} = \frac{\Delta \Psi_d(r)}{\Psi_d(r) + 1/\alpha}. \quad (115)$$

Now several test examples (pairs of explicitly given conductivities and potentials) can be constructed by translating conductivities with different parameters so that the discs where they differ from 1 do not overlap.

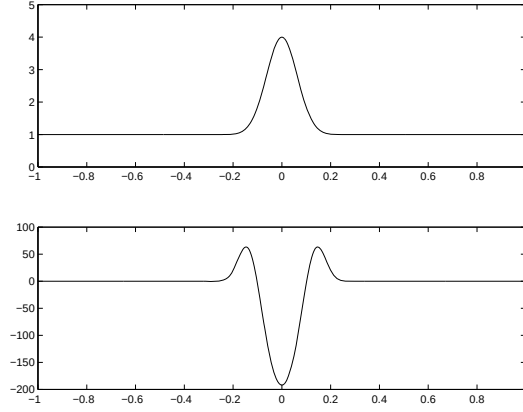


Figure 8: Profiles of the radial, smooth test conductivity γ and the corresponding potential q .

We consider here the choice $\alpha = e^8$ and $d = 1/2$; define γ and q by (114) and (115), respectively. Then $\text{supp}(\gamma - 1) = \text{supp}(q) = \overline{B(0, 1/2)}$ and both functions are radial. For plots of γ and q see Figure 8.

With the theoretical tools developed in Chapter 3 we can deduce qualitative properties of $\mathbf{t}(k)$ and $\mu(x, k)$ before any numerical computations. First of all, since $\gamma(x) = \gamma(|x|)$ and $\gamma(x) = \gamma(-\bar{x})$ for all $x \in \mathbb{R}^2$, claims (i) and (iii) of Theorem 3.19 imply that $\mathbf{t}(k)$ is radial and real-valued. Thus it is enough to compute its values only for non-negative real values $|k|$.

The functions μ have some symmetries, too, although they will not be real-valued. Formulas (78) and (80) yield $\mu(x, k_\varphi) = \mu(x_\varphi, k)$ and $\mu(x, \bar{k}) = \overline{\mu(-\bar{x}, k)}$ for all pairs $x \in \mathbb{R}^2$, $k \in \mathbb{C} \setminus 0$. We are interested only in the functions $\mu(x, |k|)$, so we can expect the involutive relation $\mu(x, |k|) = \overline{\mu(-\bar{x}, |k|)}$ from our numerically computed μ .

We note also that when solving numerically the $\bar{\partial}$ equation (11), we keep $x_0 \in \Omega$ fixed and consider $\mu(x_0, k)$ as functions of k , which is not necessarily real-valued. Then we know that there is an angle θ such that $x_0 = e^{i\theta}(-\bar{x}_0)$ and for this angle we have $\mu(x_0, k) = \overline{\mu(x_0, e^{i\theta}\bar{k})}$, and in particular the function $\mu(0, k)$ is real-valued and rotationally invariant.

Finally, the symmetries of γ can be used to reduce computational effort. Namely, the matrix elements (100) satisfy by (25)

$$\begin{aligned} a_N^{ij} &= \int_D g_{|k|}(x^{(i)} - y)q(y)v_j(y)dy = \int_D g_{|k|}(x^{(i)} - (-\bar{y}))q(-\bar{y})v_j(-\bar{y})dy \\ &= \int_D g_{|k|}(-\bar{x}^{(i')} - (-\bar{y}))q(y)v_{j'}(y)dy = \bar{a}_N^{i'j'}. \end{aligned}$$

This relation is a bit tricky to code (given a grid point $x^{(i)}$, it is not straightforward

to find a formula giving the number i' for the unique grid point $x^{(i')}$ satisfying $x^{(i)} = -\bar{x}^{(i')}$ but saves roughly half of the computations.

To ensure the reliability of the μ -computation we chose first $d = 0.5$ and seemed to have good results with $N = 29$. Then we recomputed on the square $\tilde{D}$ corresponding to $\tilde{d} = d - h$ and on the grid $\tilde{G}_{\tilde{N}}$ corresponding to $\tilde{N} = 27$. The old grid contains the new grid as a subset and we can compare the values μ_N and $\mu_{\tilde{N}}$ on the points of $G_{\tilde{N}}$. This led to relative error of magnitude 0.2%, which is reasonable since we used a fast version of the algorithm for the Faddeev Green's function. This code in turn has relative error of 0.1%.

Computing the point values of $\mu(\cdot, |k|)$ in a 27×27 grid for one k -value requires setting up the matrix (100), inverting it and applying it to the constant vector 1. This was implemented in Matlab and the code took two hours to run in a PC with Pentium MMX 166 MHz processor. See Figures 9 and 10 for plots of the functions μ on a square containing the support of q and on $\partial\Omega$.

After the μ -functions are computed, finding $\mathbf{t}(k)$ from (92) is fast. Figure 11 shows a plot of the corresponding non-physical scattering transform and the rotationally symmetric function $\hat{q}(2|k|)$.

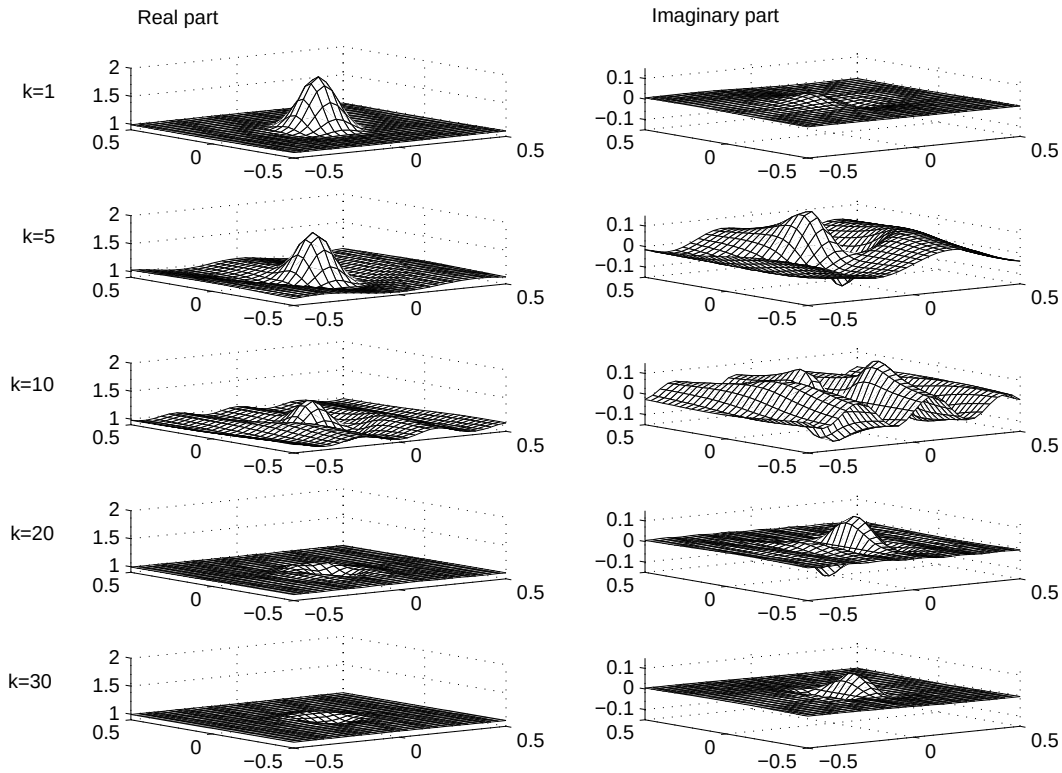


Figure 9: The functions $\mu(x, k)$ for k -values 1, 5, 10, 20 and 30. Here x ranges in the square $D = [-1/2, 1/2]^2$. Note that $\mu(x, 1)$ is very near to $\gamma^{1/2}$.

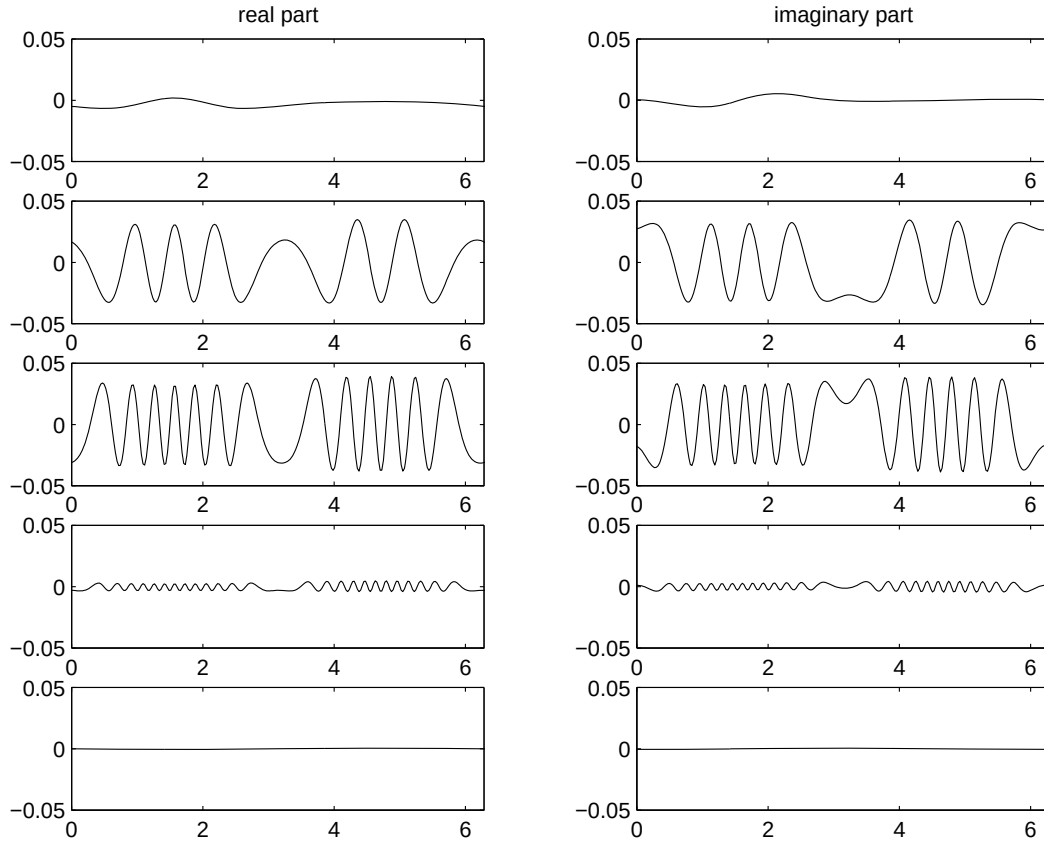


Figure 10: Boundary values on the unit circle (parametrized by angle $\varphi \in [0, 2\pi]$) of the functions $\mu(\cdot, k) - 1$ with k -values 1, 5, 10, 20 and 30.

5. Conclusion

The main purpose of this work is to provide both theoretical insight and numerical test methods for Nachman's reconstruction method. By the time of writing it is already evident that Nachman's method performs reasonably well on computer simulated data. These computations will be published elsewhere.

The results of Section 3.1 simplify working with Faddeev Green's functions in $\mathbb{R}^2$ and in Section 4.1.1 we present a fast and accurate algorithm for g_k (and consequently for G_k and H_k). In Section 3.2 we prove several theoretical results describing concrete properties of the non-physical scattering transform; these can be used to produce examples and to test numerical inversion methods. For the latter purpose we compute in Section 4.2 numerically μ and $\mathbf{t}(k)$ for a rotationally symmetric example. The methods work also for more general conductivities but are more computationally intensive then.

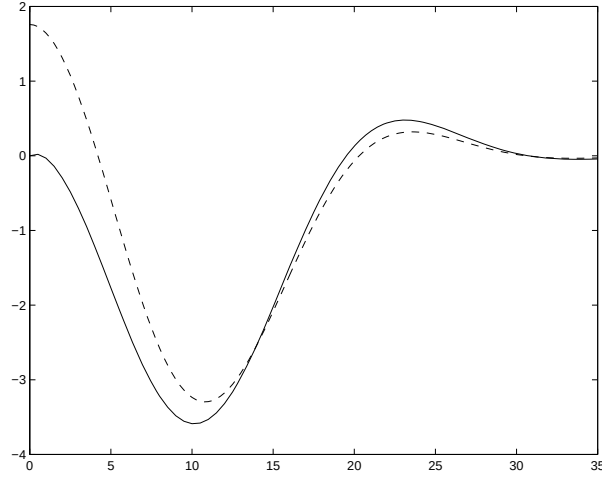


Figure 11: The non-physical scattering transform drawn with the solid line. The dashed line is the (rotationally symmetric) Fourier transform of q evaluated at $2|k|$.

Our work leads naturally to further questions and challenges about Nachman's scheme:

- Does the decay of $\mathbf{t}(k)$ imply smoothness for γ ?
- Nachman proves that $\mathbf{t}(k)$ is a continuous function. Do $\mu(x, k)$ and $\mathbf{t}(k)$ have more regularity, for instance a continuous derivative, in k ?
- In [43], Nachman does not assume that the conductivity is one near the boundary. Instead he presents a constructive procedure for recovering $\gamma|_{\partial\Omega}$ and $\partial\gamma/\partial\nu|_{\partial\Omega}$ and reduces the problem to the case $\gamma \equiv 1$ near $\partial\Omega$. Eventually the restrictions of the present work have to be dropped to produce the best possible EIT algorithm.
- However, the numerical algorithms presented in this work could find applications also elsewhere; after all, some of the deepest details of Nachman's proof come from the tradition of nonlinear evolution equations [58, 53]. If the μ -computation needs to be improved, we recommend Vainikko's FFT-based approach [54]. First of all, this would save computational effort considerably since his approach uses $\mathcal{O}(N^2 \log(N))$ arithmetical operations for one k -value whereas our code needs $\mathcal{O}(N^4)$. Secondly, convergence questions and error analysis are treated in his work elegantly using periodic pseudodifferential operators.

Note that when implementing Vainikko's scheme the derivative formulas in Section 3.1.2 are useful in computing the Fourier coefficients of the truncated Green's function $\chi_B g_k$.

- It would be interesting to consider a numerical version of the approach of Brown and Uhlmann [7] for the inversion.

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